Intrinsic-Dimension Rates and Minimax Limits for Lipschitz Wasserstein Autoencoders

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Abstract

Wasserstein autoencoders (WAEs) combine reconstruction with matching the encoded data distribution to a specified latent prior. In this study, we derive finite sample risk bounds showing how estimation accuracy depends on sample size and intrinsic dimension. Our data driven encoder and decoder classes use empirical covers of independent geometry samples, maintaining uniform Lipschitz control as approximation complexity increases. An oracle inequality separately accounts for sampling, approximation, model misspecification, and optimization errors. Under exact realization, Ahlfors regularity with a common intrinsic dimension, comparable geometry and training sample sizes, and suitable covering radii, we obtain empirical Wasserstein rates, up to optimization error: parametric order below dimension two, an additional logarithmic factor in dimension two, and dimension dependent polynomial rates above two. Euclidean reconstruction also provides Wasserstein guarantees for the generated distribution. Within an exactly realizable family obtained through smooth local transports of a fixed uniform prior, with uniform bi-Lipschitz control, a minimax lower bound matches the polynomial exponent of the corresponding upper bound, leaving a logarithmic gap in dimension two. Further results cover stochastic encoders under Hölder regularity and Gaussian latent priors through truncation. Finally, synthetic studies with 20 replications and a real data application illustrate the theoretical findings.

Keywords: Wasserstein autoencoder; intrinsic dimension; sample splitting; empirical cover; minimax lower bound; Lipschitz neural network.

1 Introduction

Wasserstein autoencoders (WAEs) are generative models that combine representation learning and data generation through an encoder–decoder architecture (Tolstikhin et al. 2018). The encoder maps observations from the data space into a typically lower dimensional latent space, while the decoder maps latent representations back to the data space. The WAE objective consists of two complementary components: a reconstruction term, which encourages the decoder to recover the original observations from their encoded representations, and a distribution matching term, which encourages the aggregate distribution of the encoded data to resemble a prescribed latent prior. The latter is measured using a chosen discrepancy between probability distributions. Once trained, the decoder can generate new observations by transforming independent samples drawn from the latent prior. Accurate reconstruction and latent distribution matching therefore both play a central role in approximating the underlying data distribution.

Beyond their generative capabilities, WAEs provide a framework for learning low dimensional representations of complex, high dimensional data and have been applied to tasks such as image and text generation, dimensionality reduction, and anomaly detection. When observations concentrate near a low dimensional set, the statistical difficulty of learning the underlying distribution may depend on its intrinsic rather than ambient dimension. Related guarantees have been established for adversarial, likelihood based, and diffusion based generative models under different structural assumptions (Huang et al. 2022; Chae et al. 2023; Oko et al. 2023; Tang and Yang 2024; Vardanyan et al. 2023).

For WAEs, Chakraborty and Bartlett (2024) established excess risk bounds by balancing neural network approximation error with generalization bounds that depend on the network

parameterization. In the Lipschitz setting, when the latent dimension does not exceed the upper Minkowski dimension d of the data support, the polynomial decay exponent in their statistical error bound can be made arbitrarily close to $1/(d+2)$, up to logarithmic factors and under their stated assumptions, including those governing latent prior sampling. Other studies examine WAEs from complementary perspectives. Husain et al. (2019) established a primal–dual relationship between generative adversarial networks and Wasserstein autoencoders, and used this connection to derive generalization bounds for autoencoder models. Qiu et al. (2025) proposed a latent Wasserstein generative adversarial framework combining Wasserstein autoencoders and adversarial learning, and established consistency of an estimator of the intrinsic dimension together with a generalization bound. More broadly, Liang (2021) investigated convergence rates, oracle inequalities, and minimax guarantees for adversarial generative models. Bodelet et al. (2026) studied nonparametric estimation of high dimensional additive latent variable models, providing theoretical guarantees on consistency and convergence rates. Chakraborty et al. (2026) established finite sample Wasserstein convergence guarantees for score based diffusion models, with rates governed by an intrinsic Wasserstein dimension under general moment conditions. These contributions illustrate different approaches to understanding the statistical properties of generative learning.

In this paper, we focus on Lipschitz Wasserstein autoencoders and analyze their statistical properties through the functional regularity of the encoder and decoder. We impose Lipschitz bounds uniformly over the candidate encoder and decoder classes and preserve these bounds throughout the lattice approximation. This separates approximation complexity from sampling error in the WAE criterion: the sampling term is controlled directly by empirical Wasserstein distances, independently of the number of lattice centers (Dudley 1969; Fournier and Guillin 2015; Weed and Bach 2019). Within this framework, we develop data driven encoder–decoder classes, finite sample guarantees that separate empirical transport, approximation, realization, and optimization errors, and minimax lower bounds under a fixed latent prior. The main contributions are described below.

1. We construct finite dimensional encoder and decoder classes using the rectified linear unit (ReLU) activation (Nair and Hinton 2010). An independent geometry split determines empirical centers, and a separate training split fits compatible lattice heights. The construction uses McShane–Whitney envelopes (McShane 1934; Whitney 1934); Lemma 5 establishes their exact ReLU representation and Lipschitz bounds independent of the number of centers. Neither the unknown data law nor values of ideal maps enter the estimator; ideal encoders and decoders serve only as proof comparators.

2. We derive a finite sample bound separating sampling error, approximation error from the empirical centers, model misspecification, and optimization error. The approximation term accounts for both the covering radius and the probability mass left uncovered. Under the lower mass conditions of Corollary 2, exact realization, comparable geometry and training sample sizes, and appropriately chosen covering radii, the bound yields the corresponding intrinsic dimensional empirical Wasserstein rates, up to optimization error. Euclidean reconstruction also gives a linear transfer to generated distribution Wasserstein error through Proposition 1.

3. We establish a minimax lower bound for distribution estimation over an exactly realizable WAE family obtained from smooth local transports of a common uniform latent prior, with uniformly controlled bi-Lipschitz constants. Applying Assouad's lemma (Assouad 1983; Pollard et al. 1997) in intrinsic dimensions of at least two and Le Cam's two point method (Pollard et al. 1997) in dimension one gives a lower bound matching the polynomial exponent of the corresponding upper bound on this family (Theorem 2). Further results cover stochastic encoders under suitable Hölder regularity conditions, and unbounded Gaussian latent priors through an increasing truncation argument.

4. Two synthetic studies, each with 20 replications, report exact evaluation of the latent transport criteria, independent cover diagnostics, and functional Lipschitz certificates. A Wisconsin illustration with 20 sample splits assesses repeatability.

The claims are conditional on the specified cover radii and Lipschitz bounds. We do not claim automatic dimension selection, polynomial time global WAE optimization, arbitrary prior realizability, or a lower bound that recovers the logarithmic factor in dimension two.

The rest of the paper is organized as follows. Section 2 defines the estimator and establishes its upper bounds and generated law consequence. Section 3 gives the fixed prior minimax result, Section 4 develops extensions, and Section 5 reports the simulations and real data illustration. The discussion and conclusion follow. The supplement contains complete proofs and computational specifications.

2 Formulation and data driven upper bound

Let P_X denote the unknown distribution of a random variable $X \in \mathcal{X} = [0,1]^d$, observed through an i.i.d. sample $\{X_i\}_{i=1}^n$ drawn from P_X. We further prescribe a latent distribution P_Z on $\mathcal{Z} = [0,1]^\ell$, chosen so that it can be readily sampled from, such as a uniform distribution. An autoencoder consists of an encoder $E : \mathcal{X} \to \mathcal{Z}$ and a decoder $G : \mathcal{Z} \to \mathcal{X}$. Given an observation X, the encoder produces the latent representation $E(X)$, which is then mapped by the decoder to the reconstruction

$$\widehat{X} = G(E(X)), \qquad X \xrightarrow{E} E(X) \xrightarrow{G} \widehat{X}.$$

The distribution of the encoded variable $E(X)$ is the pushforward $E_\# P_X$. The population reconstruction error is $\int c(x, G(E(x)))\, \mathrm{d}P_X(x)$, and the discrepancy between the encoded law and the prescribed prior is $\mathsf{W}_1(E_\# P_X, P_Z)$. Here $c : \mathcal{X} \times \mathcal{X} \to [0,\infty)$ is the reconstruction cost. For fixed $\lambda > 0$, define the *population WAE criterion*

$$\mathcal{R}_W(P_X, P_Z; G, E) = \int c\left(x, G(E(x))\right)\, \mathrm{d}P_X(x) + \lambda \mathsf{W}_1(E_\# P_X, P_Z). \tag{1}$$

We use generic laws P, Q only when explicitly discussing the same criterion for an arbitrary data/prior pair.

For the empirical data and latent distributions defined as $\widehat{P}_{X,n} = \frac{1}{n}\sum_{i=1}^{n}\delta_{X_i}$ and $\widehat{P}_{Z,m} = \frac{1}{m}\sum_{j=1}^{m}\delta_{Z_j}$, the sampled prior criterion is

$$\widehat{\mathcal{R}}_{W,n,m}(G,E) = \frac{1}{n}\sum_{i=1}^{n} c\left(X_i, G(E(X_i))\right) + \lambda \mathsf{W}_1(E_{\#}\widehat{P}_{X,n}, \widehat{P}_{Z,m}). \tag{2}$$

By minimizing (2) over pairs (E, G), we obtain the learned encoder–decoder pair denoted by $(\widehat{E}, \widehat{G})$. As is standard in statistical learning analyses, we allow for imperfect optimization of the empirical criterion. Specifically, the learned encoder–decoder pair $(\widehat{E}, \widehat{G})$ is assumed to satisfy the approximate empirical risk minimization condition:

$$\widehat{\mathcal{R}}_{W,n,m}(\widehat{G}, \widehat{E}) \leq \inf_{(G,E)\in\mathcal{G}\times\mathcal{E}} \widehat{\mathcal{R}}_{W,n,m}(G,E) + \Delta_{\mathrm{opt},n,m}, \tag{3}$$

where $\Delta_{\mathrm{opt},n,m} \geq 0$ quantifies the optimization error. This standard relaxation of exact empirical risk minimization accounts for the fact that practical training procedures, particularly for nonconvex models such as neural networks, need not attain a global minimum; exact empirical risk minimization is recovered when $\Delta_{\mathrm{opt},n,m} = 0$. While $(\widehat{G}, \widehat{E})$ is learned by minimizing the empirical criterion, the ultimate quantity of interest is its performance with respect to the underlying distributions P_X and P_Z. This is measured by the population WAE criterion $\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E})$, which evaluates the reconstruction performance on the true data distribution and the discrepancy between the distribution of the encoded data $\widehat{E}_{\#}P_X$ and the prescribed latent distribution P_Z. Our goal is therefore to quantify how the finite sample training procedure above translates into population level performance.

2.1 Observable sample split lattice estimator

In this section, we study the statistical performance of WAEs trained from finite samples and establish conditions under which their empirical performance translates into guarantees

at the population level. Related intrinsic-dimension dependent guarantees were obtained by Chakraborty and Bartlett (2024), who derive excess risk bounds by controlling the complexity of suitably chosen neural network classes. Here we take a functional approach: rather than controlling complexity through a particular network parametrization, we work with encoder and decoder classes having uniform Lipschitz regularity and construct finite approximations of these classes from empirical covers of the data and latent distributions. To state the result we make the following assumption:

Assumption C (Bounded Lipschitz reconstruction).

For $x, y, x', y' \in \mathcal{X}$, the reconstruction cost satisfies: $c(x,x) = 0,\ 0 \le c \le B_c$, and

$$|c(x,y) - c(x',y')| \le L_c \left(\|x - x'\|_2 + \|y - y'\|_2 \right).$$

We next introduce the notation and empirical cover construction.

2.1.1 Notation and empirical cover construction

The population adaptive lattice centers used in a standard approximation argument are not available to the statistician. We therefore split the observations into independent geometry and training samples,

$$\mathcal{D}_{X,\mathrm{g}} = (X_i^{\mathrm{g}})_{i \le n_{\mathrm{g}}}, \qquad \mathcal{D}_{X,\mathrm{t}} = (X_i^{\mathrm{t}})_{i \le n_{\mathrm{t}}}.$$

When prior integration is sampled, we analogously use independent prior splits of sizes m_{g} and m_{t}. For an ordered, nonempty sample $\mathcal{D} = (y_1, \dots, y_N)$, $\mathrm{Greedy}_{\varepsilon}(\mathcal{D})$ retains y_1 and then retains y_i precisely when its ℓ_∞-distance from every previously retained point exceeds ε. Consequently, every sample point lies within ε of a retained center, and distinct retained centers are more than ε apart. We use the observed sample order throughout. Set

$$C_X = \mathrm{Greedy}_{\varepsilon_X}(\mathcal{D}_{X,\mathrm{g}}), \qquad C_Z = \mathrm{Greedy}_{\varepsilon_Z}(\mathcal{D}_{Z,\mathrm{g}}). \tag{4}$$

For a known prior, C_Z may instead be a deterministic cover; for example, a Cartesian ε_Z-grid of the latent cube has quantization error at most ε_Z. For a finite nonempty C, let

$$d_\infty(u, C) = \inf_{c \in C} \|u - c\|_\infty = \min_{c \in C} \|u - c\|_\infty$$

denote the ℓ_∞-distance from u to its nearest center in C. For $r > 0$, we define

$$q_P(C) = \int d_\infty(u, C) \, \mathrm{d}P(u), \tag{5}$$

$$\eta_P(C, r) = P\{d_\infty(u, C) > r\}. \tag{6}$$

Let D_P be an upper bound on the ℓ_∞-diameter of the ambient cube ($D_P = 1$ for a unit cube). Splitting the integral over $\{d_\infty(u, C) \le r\}$ and its complement gives

$$q_P(C) \le rP\{d_\infty(u, C) \le r\} + D_P \eta_P(C, r) \le r + D_P \eta_P(C, r). \tag{7}$$

Thus, $q_P(C)$ measures how well the empirical centers represent the population without requiring the geometry observations to cover its entire support. The statistical terms arising from the independent training samples are controlled by intrinsic dimensions defined through approximate population covers. For a compactly supported law P, let $N_\varepsilon(P, \tau)$ be the smallest number of ℓ_∞-balls of radius ε covering a Borel set of mass at least $1 - \tau$. Its order one entropic and upper Wasserstein dimensions are

$$\overline{d}_1(P) = \limsup_{\varepsilon \downarrow 0} \frac{\log N_\varepsilon(P, \varepsilon)}{\log(1/\varepsilon)}, \tag{8}$$

$$d_r^\star(P) = \inf \left\{ s > 2r : \limsup_{\varepsilon \downarrow 0} \frac{\log N_\varepsilon(P, \varepsilon^{sr/(s-2r)})}{\log(1/\varepsilon)} \le s \right\}. \tag{9}$$

For $0 < r \le 1$, define the unit Hölder ball

$$\mathcal{H}_q^r(1) = \{f : \|f\|_\infty \le 1, |f(x) - f(y)| \le \|x - y\|_2^r\}.$$

Then

$$\mathbb{E}\|\widehat{P}_n - P\|_{\mathcal{H}_q^r(1)} \le C_{P,t} n^{-r/t}, \qquad t > d_r^\star(P), \tag{10}$$

with exponent slack at a critical dimension (Weed and Bach 2019; Chakraborty and Bartlett 2025). For independent $Y_1, \dots, Y_N \sim P$, write $\widehat{P}_N = N^{-1} \sum_{i=1}^N \delta_{Y_i}$. Taking $r = 1$ in (10), Kantorovich–Rubinstein duality and boundedness of the support yield

$$\mathbb{E}\mathsf{W}_1(\widehat{P}_N, P) \le C_{P,t} N^{-1/t}, \qquad t > d_1^\star(P),$$

where the finite constant absorbs the diameter dependent normalization of the Lipschitz test functions. For an Ahlfors k-regular law, $\overline{d}_1(P) = k$ and $d_1^\star(P) = k \vee 2$.

We next construct Lipschitz encoder and decoder classes from these centers. For $C = \{c_1, \dots, c_K\} \subset [0,1]^q$, cone slope L, and output dimension q', let

$$\mathcal{A}_{q'}(C, L) = \left\{ a \in [0,1]^{Kq'} : |a_{hj} - a_{h'j}| \le L\|c_h - c_{h'}\|_\infty \text{ for all } h, h', j \right\}. \tag{11}$$

Here $h, h' \in \{1, \dots, K\}$ and $j \in \{1, \dots, q'\}$. Conditional on C, $\mathcal{A}_{q'}(C, L)$ is a nonempty compact polytope, defined by finitely many linear inequalities in the height variables. Its constraints depend only on C and the prescribed slope L. The coefficient a_{hj} is the value assigned to output coordinate j at center c_h, and the constraints ensure compatibility with the prescribed Lipschitz modulus.

For every $a \in \mathcal{A}_{q'}(C, L)$, we define the clipped McShane–Whitney lattice coordinate

$$T_{a,j}(u) = \mathrm{clip}_{[0,1]}\left[\frac{1}{2} \left\{ \min_{h \le K} (a_{hj} + L\|u - c_h\|_\infty) + \max_{h \le K} (a_{hj} - L\|u - c_h\|_\infty) \right\} \right]. \tag{12}$$

where $\mathrm{clip}_{[0,1]}(v) := \min\{1, \max\{0, v\}\}$. Stacking the coordinates in (12), define

$$T_a(u) = \Big(T_{a,1}(u), \dots, T_{a,q'}(u) \Big), \qquad \mathcal{L}_{q,q'}(C, L) = \{ T_a : a \in \mathcal{A}_{q'}(C, L) \}.$$

Thus $T_a : [0,1]^q \to [0,1]^{q'}$ is a vector valued map. Each coordinate is L-Lipschitz with respect to the input ℓ_∞ norm, so $\mathrm{Lip}(T_a) \le \sqrt{q'}L$ for Euclidean norms. Set

$$\mathcal{E}(C_X) = \mathcal{L}_{d,\ell}(C_X, L_E), \qquad \mathcal{G}(C_Z) = \mathcal{L}_{\ell,d}(C_Z, L_G). \tag{13}$$

In particular, these encoder and decoder classes have Euclidean Lipschitz bounds $K_E = \sqrt{\ell}L_E$ and $K_G = \sqrt{d}L_G$, independently of their numbers of centers.

The heights are free optimization variables, and every candidate has a common functional Lipschitz bound independent of the number of centers. Finite min–max trees also give exact ReLU implementations with $O(K(q+q'))$ nonzero parameters and logarithmic depth.

Conditional on the geometry samples, the resulting classes are fixed and the encoder–decoder pair is learned using only the independent training samples through

$$\widehat{\mathcal{R}}_{W,\mathrm{t}}(G,E) = \frac{1}{n_\mathrm{t}} \sum_{i=1}^{n_\mathrm{t}} c(X_i^\mathrm{t}, G(E(X_i^\mathrm{t}))) + \lambda \mathsf{W}_1(E_\# \widehat{P}_{X,\mathrm{t}}, \widehat{P}_{Z,\mathrm{t}}), \tag{14}$$

replacing $\widehat{P}_{Z,\mathrm{t}}$ by P_Z under exact prior integration. We allow the learned pair $(\widehat{G}, \widehat{E})$ to be a measurable Δ_{opt}-approximate minimizer.

Finally, for fixed Lipschitz reference maps E_0, G_0, we define the reconstruction and aggregate matching defects

$$b^0_{\mathrm{rec}} = \int c(x, G_0(E_0(x)))\, \mathrm{d}P_X(x), \qquad b^0_{\mathrm{agg}} = \mathsf{W}_1(E_{0\#}P_X, P_Z). \tag{15}$$

These terms quantify the misspecification of the reference pair and vanish under exact realization.

With the empirical cover construction and training procedure in place, we now state the main finite sample guarantee. Fix Lipschitz reference maps $E_0 : \mathcal{X} \to \mathcal{Z}$ and $G_0 : \mathcal{Z} \to \mathcal{X}$. They may depend on the population laws but are used only as proof comparators. Let $n_\mathrm{g}, n_\mathrm{t} \ge 1$. When both prior splits are sampled, take $m_\mathrm{g}, m_\mathrm{t} \ge 1$, with all four splits mutually

independent. Exact prior integration removes the prior training sample, and a deterministic prior cover removes the prior geometry sample. Write $\mathcal{D}_{\mathrm{g}} = (\mathcal{D}_{X,\mathrm{g}}, \mathcal{D}_{Z,\mathrm{g}})$ when both covers are random, and $\mathcal{D}_{\mathrm{g}} = \mathcal{D}_{X,\mathrm{g}}$ when the prior cover is deterministic. Below, $\mathbb{E}_{\mathrm{t}}[\cdot \mid \mathcal{D}_{\mathrm{g}}]$ averages over the independent training samples and any independent optimization randomness.

Theorem 1 (Sample split empirical cover oracle inequality)**.** *Suppose Assumption C holds. Choose* $L_E \geq \sqrt{d}\,\mathrm{Lip}(E_0)$ *and* $L_G \geq \sqrt{\ell}\,\mathrm{Lip}(G_0)$*. For* $t_X > d_1^\star(P_X)$ *and* $t_Z > d_1^\star(P_Z)$*, every measurable* Δ_{opt}*-approximate minimizer satisfies, conditionally on the geometry samples,*

$$\mathbb{E}_{\mathrm{t}}\Big[\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \mid \mathcal{D}_{\mathrm{g}}\Big] \leq b^0_{\mathrm{rec}} + C_{\mathrm{agg}} b^0_{\mathrm{agg}} + C\{q_{P_X}(C_X) + q_{P_Z}(C_Z) + n_{\mathrm{t}}^{-1/t_X} + m_{\mathrm{t}}^{-1/t_Z}\} + \mathbb{E}_{\mathrm{t}}(\Delta_{\mathrm{opt}} \mid \mathcal{D}_{\mathrm{g}}), \tag{16}$$

where $C_{\mathrm{agg}} = \lambda + L_c\{\sqrt{d}L_G + \mathrm{Lip}(G_0)\}$ *is admissible. Exact prior integration deletes* m_{t}^{-1/t_Z}*; a deterministic prior cover leaves its nonrandom* $q_{P_Z}(C_Z)$*. If* $c(x,y) = \|x - y\|_2$*, the right side of Equation* (16)*, multiplied by* $\max\{1, \sqrt{d}L_G/\lambda\}$*, bounds*

$$\mathbb{E}_{\mathrm{t}}\Big[\mathsf{W}_1(\widehat{G}_{\#}P_Z, P_X) \mid \mathcal{D}_{\mathrm{g}}\Big].$$

Theorem 1 is an upper bound result for a specified random sieve. Sample splitting makes that sieve fixed during the training sample analysis. It neither assumes that the empirical centers cover the support nor hides global WAE optimization: the latter remains in Δ_{opt}. Theorem 1 answers the following question: after the geometry sample has fixed the candidate classes, how well does the WAE learned from an independent finite training sample perform on the population laws? Its right-hand side separates four sources of error.

- **Comparator defects.** b^0_{rec} and b^0_{agg} measure imperfect reconstruction and latent matching by the reference pair E_0, G_0. They vanish under exact realization.
- **Quantization.** $q_{P_X}(C_X)$ and $q_{P_Z}(C_Z)$ measure average distances from population draws to the geometry centers. They control the cost of approximating the reference

maps by compatible lattices.

- **Sampling.** The training sample terms arise from empirical Wasserstein errors for the data and prior. The prior term disappears under exact prior integration.
- **Optimization.** $\mathbb{E}_{\mathrm{t}}[\Delta_{\mathrm{opt}} \mid \mathcal{D}_{\mathrm{g}}]$ retains any failure to minimize the empirical criterion globally. It is the actual empirical suboptimality gap or a valid upper bound; the theorem does not itself bound this computational error.

The quantization terms are fixed inside the conditional expectation; averaging over the geometry samples subsequently yields the unconditional bounds.

The cover cost can be averaged exactly. For iid $Y_1, \ldots, Y_N \sim P$, define

$$\nu_P(N) = \int\int_0^{D_P} \{1 - P(B_\infty(u, r))\}^N \, \mathrm{d}r \, \mathrm{d}P(u). \tag{17}$$

If $C = \mathrm{Greedy}_\varepsilon(Y_{1:N})$, then

$$\mathbb{E}q_P(C) \le \varepsilon + \nu_P(N). \tag{18}$$

The preceding bound allows the reference pair to have nonzero reconstruction and aggregate matching defects. We now impose exact realization.

Assumption LR (Lipschitz WAE realization).

There exist fixed Lipschitz maps $E_0 : \mathcal{X} \to \mathcal{Z}$ and $G_0 : \mathcal{Z} \to \mathcal{X}$ such that

$$E_{0\#}P_X = P_Z, \qquad G_0(E_0(x)) = x \quad P_X\text{-almost surely.}$$

On suitable full measure sets, these maps form a bi-Lipschitz chart, so this is a substantive structural condition. Under Assumption LR, $b^0_{\mathrm{rec}} = b^0_{\mathrm{agg}} = 0$. Theorem 1 therefore leaves only quantization, sampling and optimization terms, as stated in Corollary 1.

Corollary 1 (Exact realization)**.** *Under Assumption LR, take its E_0, G_0 in Theorem 1. Then $b^0_{\mathrm{rec}} = b^0_{\mathrm{agg}} = 0$, so only empirical transport, quantization, and optimization terms remain.*

To turn Theorem 1 into a rate in the sample sizes, we average over the geometry samples and bound the expected quantization errors using (17)–(18). An additional lower mass condition controls the probability that a neighborhood of a population draw contains no geometry observation. It also bounds the empirical Wasserstein errors; substituting both estimates into Theorem 1 gives Corollary 2.

For each $P \in \{P_X, P_Z\}$, assume that there exist constants $a_P > 0$, $r_P > 0$ and $k_P > 0$ such that

$$P\{B_\infty(u, r)\} \geq a_P r^{k_P}, \qquad u \in \operatorname{supp} P, \quad 0 < r \leq r_P. \tag{19}$$

These constants describe the population law: a_P is the lower mass coefficient, r_P is the range of radii over which the inequality holds, and k_P is its exponent. For an Ahlfors k-regular law one may take $k_P = k$. For a general law, a lower mass exponent need not be a uniquely identified intrinsic dimension. The condition is a stated assumption, not an estimated input obtained from the observed centers.

There is then a finite constant $C_{P,\mathrm{q}}$, depending only on a_P, r_P, k_P and the ambient diameter and not on N, such that

$$\nu_P(N) \leq C_{P,\mathrm{q}} N^{-1/k_P}. \tag{20}$$

Indeed, with $\bar{r}_P = \min\{r_P, D_P\}$, the inequality $(1-u)^N \leq e^{-Nu}$ gives

$$\nu_P(N) \leq \Gamma(1 + 1/k_P)(a_P N)^{-1/k_P} + D_P e^{-a_P N \bar{r}_P^{k_P}}.$$

The second term can be absorbed into a constant multiple of N^{-1/k_P}. We use $C_{P,\mathrm{W}}$ for the possibly different constant in the empirical Wasserstein bound; it may also depend on the ambient dimension through norm conversion.

We define

$$\rho_{N,k} = \begin{cases} N^{-1/2}, & 0 < k < 2, \\ N^{-1/2} \log(1+N), & k = 2, \\ N^{-1/k}, & k > 2. \end{cases} \tag{21}$$

The same lower mass condition implies the dimension dependent empirical upper bound

$$\mathbb{E}\mathsf{W}_1(\widehat{P}_N, P) \le C_{P,\mathrm{W}} \rho_{N,k_P}. \tag{22}$$

Thus, averaged over independent samples of size N, the empirical distribution lies within a constant multiple of ρ_{N,k_P} of P in Wasserstein–1 distance. This is an upper bound; it does not assert that every law satisfying the condition has this exact convergence rate.

Corollary 2 (Regular law rate)**.** *Under the hypotheses of Theorem 1 and the preceding lower mass conditions,*

$$\begin{aligned} \mathbb{E}\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \le b^0_{\mathrm{rec}} + C_{\mathrm{agg}} b^0_{\mathrm{agg}} + \mathbb{E}\Delta_{\mathrm{opt}} \\ + C\{\rho_{n_{\mathrm{t}},k_X} + \rho_{m_{\mathrm{t}},k_Z} + \varepsilon_X + n_{\mathrm{g}}^{-1/k_X} + \varepsilon_Z + m_{\mathrm{g}}^{-1/k_Z}\}. \end{aligned} \tag{23}$$

The displayed formula uses a sampled greedy prior cover. Exact prior integration removes the prior training term; with a deterministic prior cover, replace $\varepsilon_Z + m_{\mathrm{g}}^{-1/k_Z}$ by $q_{P_Z}(C_Z)$. If both laws are Ahlfors k-regular, the four sample sizes are comparable to N, $\varepsilon_X + \varepsilon_Z = O(\rho_{N,k})$, and the comparator is exact, then

$$\mathbb{E}\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \le C\rho_{N,k} + \mathbb{E}\Delta_{\mathrm{opt}}. \tag{24}$$

Moreover, for positive tolerances below the regularity radii, $|C_X| \le \min\{n_{\mathrm{g}}, C\varepsilon_X^{-k_X}\}$; the analogous prior bound holds for a sampled greedy prior cover.

For independent $Y_1, \ldots, Y_N \sim P$, a greedy ε-cover $C = \mathrm{Greedy}_\varepsilon(Y_{1:N})$ covers every

observed point at radius ε, but need not cover the entire population support. Regions of positive probability may contain no sampled observation. The quantity $\eta_P(C, s)$ defined in (6) measures the population mass farther than s from the retained centers. The following bound controls this missed mass after enlarging the radius from ε to $\varepsilon + r$.

For $0 < \delta < 1$ and $0 < r \le r_P$, the same lower-mass condition gives

$$\mathbb{P}\Big\{\eta_P(C, \varepsilon + r) > \delta^{-1} e^{-a_P N r^{k_P}}\Big\} \le \delta. \tag{25}$$

The outer probability in (25) is over the independent geometry sample $Y_1, \ldots, Y_N \sim P$. The inner quantity $\eta_P(C, \varepsilon + r)$ is the population mass missed by the resulting random center set.

For $0 < \delta < 1$, choose

$$r = \left\{\frac{2\log(N/\delta)}{a_P N}\right\}^{1/k_P},$$

provided that $r \le r_P$. Then, with probability at least $1 - \delta$ over the geometry sample,

$$\eta_P(C, \varepsilon + r) \le \delta/N^2, \qquad q_P(C) \le \varepsilon + r + D_P\delta/N^2. \tag{26}$$

The logarithmic factor is the cost of this particular near coverage guarantee; it is absent from the expected integrated quantization bound. The result does not assert complete support coverage. The construction prescribes $\varepsilon_X, \varepsilon_Z, L_E, L_G$ and does not automatically select the intrinsic dimension or the regularity radius.

2.2 Generated law consequence

The preceding results control the population WAE criterion. When reconstruction is measured by Euclidean distance, the coupling $(X, G(E(X)))$ bounds the transport cost from P_X to $(G \circ E)_{\#}P_X$. Moreover, Wasserstein–1 distance contracts under a Lipschitz decoder. Combining these facts gives the following guarantee for the generated law $G_{\#}P_Z$.

Proposition 1 (Generated law transfer)**.** *If* $c(x,y) = \|x-y\|_2$ *and* $\mathrm{Lip}(G) \le K_G$ *for Euclidean norms, then*

$$\begin{aligned}\mathsf{W}_1(G_{\#}P_Z, P_X) &\le \int \|x - G(E(x))\|_2 \,\mathrm{d}P_X(x) + K_G \mathsf{W}_1(E_{\#}P_X, P_Z) \\ &\le \max\{1, K_G/\lambda\}\mathcal{R}_W(P_X, P_Z; G, E). \end{aligned} \tag{27}$$

Here K_G is a Euclidean Lipschitz bound. For the decoder lattice in Theorem 1, one may take $K_G = \sqrt{d}L_G$, where L_G is the coordinate cone slope.

The proof couples X with $G(E(X))$, contracts Wasserstein distance under G, and applies the triangle inequality. For squared Euclidean reconstruction, Cauchy–Schwarz instead gives

$$\mathsf{W}_1(G_{\#}P_Z, P_X) \le \left\{\int \|x - G(E(x))\|_2^2 \,\mathrm{d}P_X(x)\right\}^{1/2} + K_G \mathsf{W}_1(E_{\#}P_X, P_Z).$$

Thus the linear transfer from the WAE criterion in Proposition 1 is stated specifically for Euclidean, rather than squared Euclidean, reconstruction.

The supplement records a population cover oracle benchmark that separates entropic approximation dimension from empirical transport dimension. It is useful for comparison with earlier work but is not the estimator analyzed in Theorem 1: the empirical cover construction above is the paper's central statistical procedure.

3 A Fixed Prior WAE Minimax Theorem

When both laws are Ahlfors k-regular, the geometry and training sample sizes are comparable to N, and the cover radii are $O(\rho_{N,k})$, combining Corollary 2 with Proposition 1 yields a generated law upper bound of order $\rho_{N,k}$, apart from comparator and optimization errors. This transfer uses Euclidean reconstruction. We now compare this polynomial rate with a lower bound inside an exact WAE model sharing one fixed prior. For this model we also construct deterministic grid WAE classes, chosen independently of the unknown member

P, that attain a uniform upper bound of the same order. Equation (22) is the empirical transport ingredient in this argument.

The construction below uses the prescribed prior $U_k = \mathrm{Unif}([0,1]^k)$ throughout. Each data law is a smooth bi-Lipschitz push-forward of U_k, with common Lipschitz constants for the realizing encoder and decoder.

To obtain a sufficiently rich family while retaining uniform Lipschitz control, we partition $[0,1]^k$ into cells of side length $h = m^{-1}$ and introduce an independent smooth local perturbation within each cell. For $d \geq k$, define the coordinate embedding and projection by

$$\iota_k(z) = (z_1, \ldots, z_k, \quad \underbrace{0, \ldots, 0}_{d-k \text{ coordinates}} \quad), \qquad \pi_k(x) = (x_1, \ldots, x_k).$$

Thus 0_{d-k} denotes the zero vector in $\mathbb{R}^{d-k}$. When $d = k$, no zero coordinates are appended and both maps are identities.

We choose a nonconstant $\psi \in C_c^\infty((1/4, 3/4)^k)$, and we put $V = \nabla\psi$. For each integer $m \geq 1$, let $h = m^{-1}$, index the regular cells by $j \in \{0, \ldots, m-1\}^k$, and set

$$T_{m,\alpha}(x) = x + \sum_j \alpha_j h V\left(\frac{x - hj}{h}\right), \qquad \alpha \in [-a_0, a_0]^{m^k}, \tag{28}$$

where every summand is extended by zero outside its cell and $a_0 > 0$ is a sufficiently small fixed constant. Let $\mathcal{T}_k = \bigcup_{m \geq 1} \{T_{m,\alpha} : \alpha \in [-a_0, a_0]^{m^k}\}$. The perturbation amplitude is proportional to h. Consequently, every resulting map T is C^∞ and both $\mathrm{Lip}(T)$ and $\mathrm{Lip}(T^{-1})$ are bounded independently of m and α. Define the single, sample size independent family

$$\Lambda_k^{\mathrm{fp}} = \{\iota_{k\#}(T_\# U_k) : T \in \mathcal{T}_k\}. \tag{29}$$

Every member of Λ_k^{fp} is exactly realizable using the same prior U_k. For $P = \iota_{k\#}(T_\# U_k)$, take

$$G_T(z) = \iota_k\{T(z)\}, \qquad E_T(x) = T^{-1}\{\pi_k(x)\}. \tag{30}$$

Then $E_{T\#}P = U_k$ and $G_T\{E_T(X)\} = X$ almost surely. The encoder and decoder Lipschitz constants are bounded uniformly over the family because the coordinate embedding and projection are nonexpansive and the bounds for T, T^{-1} are uniform. Thus every law in Equation (29) belongs to one exact WAE realizable class with the same prior and common encoder–decoder Lipschitz bounds.

Theorem 2 (Fixed prior WAE minimax rate)**.** *Fix an integer $1 \le k \le d$ and $\lambda > 0$. The constant a_0 in the preceding construction can be chosen so that the following statements hold for all $n \ge 1$.*

Lower bound. *There exists a constant $c_k > 0$ such that*

$$\inf_{\widehat{P}_n} \sup_{P \in \Lambda_k^{\mathrm{fp}}} \mathbb{E}_P \mathsf{W}_1(\widehat{P}_n, P) \ge c_k n^{-1/(k \vee 2)}. \tag{31}$$

The infimum ranges over all measurable probability distribution estimators based on n observations, including estimators with independent auxiliary randomization. Hence the bound applies to WAE generated distributions and remains valid when arbitrary independent samples from the known prior U_k are supplied.

Upper bound. *Use Euclidean reconstruction, a Wasserstein–1 latent penalty, and exact prior integration. There are deterministic grid lattice encoder and decoder classes, with uniform Lipschitz bounds and chosen independently of the unknown $P \in \Lambda_k^{\mathrm{fp}}$, such that a measurable exact empirical minimizer obeys*

$$\sup_{P \in \Lambda_k^{\mathrm{fp}}} \mathbb{E}_P \mathsf{W}_1(\widehat{G}_{n\#} U_k, P) \le C_k \rho_{n,k}, \qquad \rho_{n,k} = \begin{cases} n^{-1/2}, & k = 1, \\ n^{-1/2} \log(1+n), & k = 2, \\ n^{-1/k}, & k \ge 3. \end{cases} \tag{32}$$

For a measurable $\Delta_{\mathrm{opt,n}}$-approximate minimizer, the right-hand side becomes

$$C_k \rho_{n,k} + C \sup_{P \in \Lambda_k^{\mathrm{fp}}} \mathbb{E}_P \Delta_{\mathrm{opt,n}}.$$

The upper bound constants may depend on the fixed dimensions, λ, and the prescribed Lipschitz bounds, but not on n or the unknown member P. The corresponding network size is $O(\rho_{n,k}^{-s})$ for each fixed $s > k$. These classes use grids on the known coordinate support and latent cube; the result establishes existence of an empirical minimizer and does not provide a polynomial time optimization guarantee.

For $k \geq 2$, the proof uses the subfamily $\alpha_j = a\theta_j$ with fixed small $a > 0$ and $\theta_j \in \{-1, 1\}$. Two adjacent vertices differ in one cell. Their one sample KL and squared Hellinger distances are both $O(a^2 h^k)$. A sum of disjoint local Kantorovich potentials is globally one-Lipschitz and proves the nonlocal separation

$$\mathsf{W}_1(P_\theta, P_{\theta'}) \geq c a h^{k+1} H(\theta, \theta'). \tag{33}$$

Taking $m^k \asymp n$ gives $cn^{-1/k}$ for $k \geq 2$. For $k = 1$, the fixed continuum class in Equation (28) contains the two amplitudes $\alpha = \pm cn^{-1/2}$; Le Cam's inequality gives the parametric lower bound. For $k \geq 2$, a Varshamov–Gilbert subcode and Fano's inequality give the same conclusion. Complete density, divergence, and dual separation calculations appear in the fixed prior minimax appendix of the online supplement.

The upper classes use deterministic regular grids, rather than unknown law adaptive centers: one grid lies on the known coordinate plane supporting every data law and one lies on the latent cube; all admissible lattice heights are free. Their cardinalities are $O(\varepsilon^{-k})$, and the common Lipschitz constants give uniform approximation over the whole fixed prior family. If $X_i = \iota_k\{T(Z_i)\}$, then

$$\mathsf{W}_1(P_n, P) \leq \mathrm{Lip}(T) \mathsf{W}_1(U_{k,n}, U_k).$$

Combining the cube empirical transport rate with the lattice approximation and generated law transfer results gives Equation (32). At fixed architecture size the centers are deterministic and the free lattice heights range over a compact set; the empirical objective is continuous, so an exact empirical minimizer exists; this is an existence statement rather than a computational guarantee.

Interpretation. Theorem 2 establishes exponent optimality within a nontrivial exact WAE model with a fixed prior. It is exact at the polynomial level for $k = 1$ and $k \geq 3$, and exponent level at the critical dimension $k = 2$. The logarithmic factor at $k = 2$ is not settled.

The smoothness qualification matters. Each hard transport is C^∞, but its higher derivatives are not bounded uniformly as the cell scale tends to zero. The theorem is therefore a lower bound for the uniformly bi-Lipschitz model used by the lattice upper theorem, not for a fixed C^r or Sobolev ball with $r > 1$. It also does not provide a polynomial time method for attaining zero optimization error. These are boundaries of the result, not omitted consequences of the proof.

4 Extensions Beyond Exact Deterministic Realizability

We record three extensions separating the core sampling argument from exact one chart realizability; full statements, variants, and proofs are in the supplement.

4.1 Approximate Lipschitz realizability

Exact realizability is not needed for the upper inequality. For any fixed Lipschitz comparators E_0, G_0, Theorem 1 retains the reconstruction defect b^0_{rec} and aggregate posterior defect b^0_{agg} in Equation (15). These are comparator bias terms, not quantities supplied to the algorithm: the empirical centers and compatible height polytopes are unchanged. Corollary 1 is the special case in which both defects vanish.

4.2 Wasserstein–Hölder stochastic encoders

For a kernel $Q(\mathrm{d}z \mid x)$, let $Q_x = Q(\cdot \mid x)$ and $(Q_{\#}P)(A) = \int Q(A \mid x)\mathrm{d}P(x)$. Define

$$\mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; G, Q) = \iint c\{x, G(z)\}\, Q(\mathrm{d}z \mid x)\mathrm{d}P_X(x) + \lambda \mathsf{W}_1(Q_{\#}P_X, P_Z). \tag{34}$$

Assumption SU requires fixed $0 < a_Q, a_G \leq 1$ and $L_Q, L_G < \infty$ such that every candidate satisfies $\mathsf{W}_1(Q_x, Q_{x'}) \leq L_Q \|x - x'\|_2^{a_Q}$ and $\|G(z) - G(z')\|_2 \leq L_G \|z - z'\|_2^{a_G}$. The empirical criterion integrates each Q_{X_i} exactly. Here L_Q and L_G are the functional Hölder moduli for this extension; in particular, this local L_G is not the coordinate cone slope used to define the earlier lattice classes. The candidate classes are fixed independently of the observations used to fit them.

Theorem 3 (Oracle rate for stochastic encoders)**.** *Under Assumptions C and SU, put $\gamma = a_Q a_G$. Let $\mathcal{G}_n$ and $\mathcal{Q}_n$ be fixed nonempty candidate classes. A measurable $\Delta_{\mathrm{opt,n,m}}$-approximate minimizer over $\mathcal{G}_n \times \mathcal{Q}_n$ obeys, for $t_X > d_\gamma^\star(P_X)$ and $t_Z > d_1^\star(P_Z)$,*

$$\mathbb{E}\mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; \widehat{G}, \widehat{Q}) \leq \inf_{G \in \mathcal{G}_n,\, Q \in \mathcal{Q}_n} \mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; G, Q) + \mathbb{E}\Delta_{\mathrm{opt,n,m}} + C\{n^{-\gamma/t_X} + m^{-1/t_Z}\}. \tag{35}$$

The latent term is absent under exact prior integration. If decoders are uniformly Lipschitz and $c(x, y) = \|x - y\|_2$, then

$$\mathsf{W}_1(G_{\#}P_Z, P_X) \leq \iint \|x - G(z)\|_2 Q(\mathrm{d}z \mid x)\mathrm{d}P_X(x) + \mathrm{Lip}(G)\mathsf{W}_1(Q_{\#}P_X, P_Z). \tag{36}$$

Finite conditional sampling adds an indexed empirical process; the theorem does not absorb that Monte Carlo error without a reparameterization or entropy argument.

4.3 A qualified Gaussian prior truncation result

Let $\Gamma_\ell = N(0, I_\ell)$, let π_R clip coordinates to $[-R, R]$, and put $\Gamma_{\ell,R} = \pi_{R\#}\Gamma_\ell$. Then

$$\tau_\ell(R) = \mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R}) \le \sqrt{2/\pi}\, \ell e^{-R^2/2}. \tag{37}$$

Assumption TR(κ) requires reference maps on each truncated cube, with defects $b_{\mathrm{rec}}(R)$ and $b_{\mathrm{agg}}(R)$, and bounds every constant in the approximation, sampling, oracle, and generation chain by $K_R \le C(1+R)^\kappa$; decoders remain globally Lipschitz on $\mathbb{R}^\ell$.

Corollary 3 (Gaussian prior by increasing truncation)**.** *Let $r_{n,m} = n^{-1/t_X} + m^{-1/t_Z}$, where $t_X > d_1^\star(P_X)$ and $t_Z > \ell \vee 2$. Under Assumptions C and TR(κ), with Euclidean reconstruction $c(x,y) = \|x-y\|_2$, for each $R \ge R_0$ take the population cover lattice classes of Theorem 6, after affine rescaling of $[-R,R]^\ell$ to the unit cube. These classes are existential and may use population dependent centers; this corollary does not assert the same rate for the sample split empirical cover classes without their additional quantization terms. Trained against $\Gamma_{\ell,R}$ at tolerance scale $r_{n,m}$ as specified in Proposition 4, they satisfy*

$$\mathbb{E}\mathsf{W}_1(\widehat{G}_{n,m,R\#}\Gamma_\ell, P_X) \le C(1+R)^\kappa\{b_{\mathrm{rec}}(R) + b_{\mathrm{agg}}(R) + r_{n,m} + \mathbb{E}\Delta_{\mathrm{opt,n,m,R}} + \tau_\ell(R)\}. \tag{38}$$

For $N = n \wedge m \ge 3$, choose $R_N = R_0 \vee \{2A\log N\}^{1/2}$ with $A > \max\{1/t_X, 1/t_Z\}$. Then

$$\mathbb{E}\mathsf{W}_1(\widehat{G}_{n,m,R_N\#}\Gamma_\ell, P_X) \le C(\log N)^{\kappa/2}\{b_{\mathrm{rec}}(R_N) + b_{\mathrm{agg}}(R_N) + r_{n,m} + \mathbb{E}\Delta_{\mathrm{opt,n,m,R_N}}\}. \tag{39}$$

Exact prior integration uses $r_n = n^{-1/t_X}$ and $N = n \ge 3$, omits the m term, and requires only $A > 1/t_X$.

The corollary neither guarantees vanishing defects nor asserts TR(κ) for an arbitrary law. It quantifies the tail and stability cost when the compact problems are controlled along the truncation sequence.

5 Theorem matched experiments and a real data illustration

The experiments ask narrower questions than a benchmark study: does explicit functional Lipschitz control behave as claimed; can one measure the exact latent transport term; and does the realizable sieve improve with sample size when optimization is not a confounder? All reported configurations were fixed before the frozen run. There are 20 fixed replications per synthetic configuration and 20 repeated stratified real data splits; no failed fit was removed, and all raw results and seedwise summaries are included in the supplement.

5.1 Exactly realizable single chart design

Let $U \sim \mathrm{Unif}[0,1]$ and $X = T_0(U) \in [0,1]^{10}$, where

$$T_{0,1}(u) = u,$$
$$T_{0,j}(u) = 0.5 + 0.18\, s_j\{2\pi u + 0.37(j-1)\} + 0.12u, \qquad j = 2, \ldots, 10,$$

where s_j is sine for even j and cosine for odd j. The encoder $E_0(x) = x_1$ and decoder $G_0 = T_0$ satisfy Assumption LR with the fixed uniform prior. We use $n \in \{128, 256, 512, 1024\}$ and an independent test law in every replication.

Four neural estimators share a 24-unit one hidden layer tanh encoder and decoder, the same training sample PCA initialization, identical parameter count, 500 full batch Adam updates, and the Euclidean reconstruction cost. They differ only in regularization: an autoencoder (AE), exact one-dimensional W_1 aggregate matching (WAE–W1), a 32-feature random Fourier MMD approximation (WAE–MMD), and WAE–W1 with spectral projection after every update (Lip–WAE–W1). Tanh and clipping are one-Lipschitz; the projected matrix norms certify encoder and decoder bounds of 2 and 30. The exact one-dimensional latent discrepancy is evaluated by integrating the empirical quantile function against the

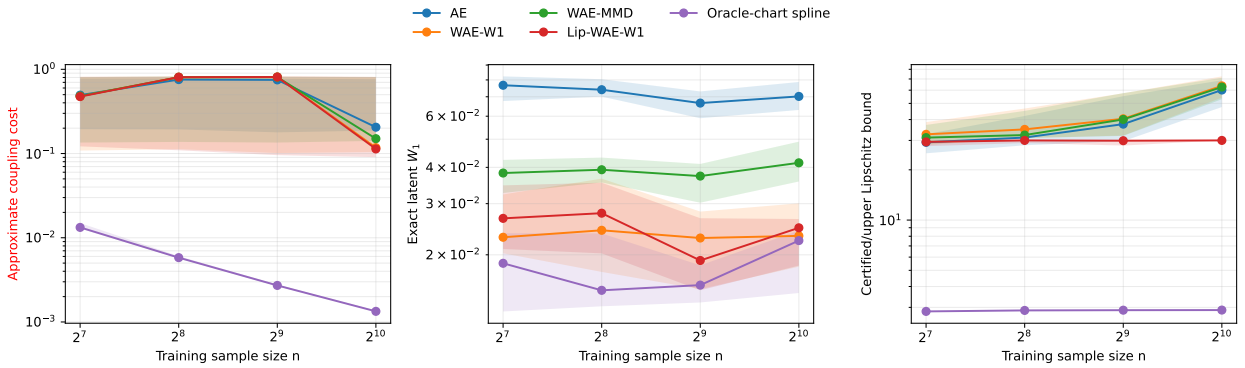


Figure 1: Exactly realizable experiment. Curves show medians and interquartile ranges across 20 replications. Left: midpoint approximation to a common latent coupling upper bound on population W_1. Center: exact latent W_1. Right: decoder Lipschitz certificate or matrix norm upper bound. The oracle chart spline is a theorem sieve diagnostic; the four neural methods are architecture- and update matched.

uniform quantile function, not by sliced transport.

A fifth estimator, labelled *oracle chart spline*, fixes the known encoder x_1 and fits a piecewise linear decoder on $O(\sqrt{n})$ knots. Adjacent knot values are projected onto a Euclidean slope ball. It is not an architecture matched competitor: its purpose is to isolate a constructive Lipschitz sieve from neural optimization and chart estimation, as motivated by Theorem 1.

We report test reconstruction, exact latent W_1, equal weight ambient empirical W_1 computed by the Hungarian algorithm, and matrix product Lipschitz bounds. The ambient empirical metric has a visible finite evaluation floor. We therefore also evaluate the fitted decoder and T_0 on the same dense uniform latent grid. Their mean Euclidean separation is a midpoint approximation to a common latent coupling upper bound on population W_1, rather than a certified population upper bound or the optimal transport distance. Log–log slopes are first fitted within replication and then summarized across the 20 replications.

The results separate approximation from optimization. The oracle chart spline has mean seedwise slopes -1.119 for the common latent coupling cost (95% normal interval -1.142 to -1.097) and -1.125 for reconstruction. These slopes are descriptive and are not estimates of a minimax exponent; the noiseless one-dimensional curve is smoother than the general

Lipschitz class. Lip–WAE–W1 respects its decoder certificate (mean 29.75 at the largest sample size), whereas the matched unconstrained WAE–W1 matrix norm bound is 63.81. Their corresponding exact latent W_1 means are 0.025 and 0.024. The constrained neural fit need not dominate reconstruction: imposing a certificate creates an observable approximation–optimization tradeoff. This is consistent with, but does not prove, the theorem's requirement that optimization error and approximation error both be controlled. The full method by method table, including Monte Carlo standard errors, is in the supplement.

5.2 Sample split empirical cover diagnostic

We separately implement a uniformly Lipschitz, coordinate specific slope variant of the empirical cover construction in Theorem 1 on fixed, smooth, injective charts of dimensions $k = 1$ and $k = 2$ in $[0,1]^{10}$. For each total size $n \in \{128, 256, 512, 1024\}$, exactly half the observations form $\mathcal{D}_{X,\mathrm{g}}$ and half form $\mathcal{D}_{X,\mathrm{t}}$. The geometry split is used only for the greedy ℓ_∞ cover C_X. The training split fits a capped PCA coordinate system; its values at C_X are projected onto the compatible height polytope. Decoder heights are training sample cell means, with an empty cell copying the nearest nonempty training derived height. A deterministic tensor grid forms C_Z for the known uniform prior. Both maps use the coordinatewise Whitney–McShane envelope form of the theorem, with cone slopes computed from the fitted compatible heights and bounded by fixed coordinate caps. They carry recorded global Lipschitz certificates. No population chart, held out observation, or intrinsic dimension estimator enters the fitted heights. The value of k is supplied by each synthetic design and is not estimated or adaptively selected.

Independent samples of size 1,024 estimate q_X, q_Z, the mass beyond the prescribed cover radius, and its one-sided Wilson upper limit. The encoder and decoder Lipschitz certificates are computed exactly from the fitted lattice heights. For $k = 1$, latent discrepancy is exact empirical to continuous uniform W_1; for $k = 2$, it is exact equal weight Hungarian W_1 to an independent same size prior sample, with sliced W_1 retained only as a secondary diagnostic.

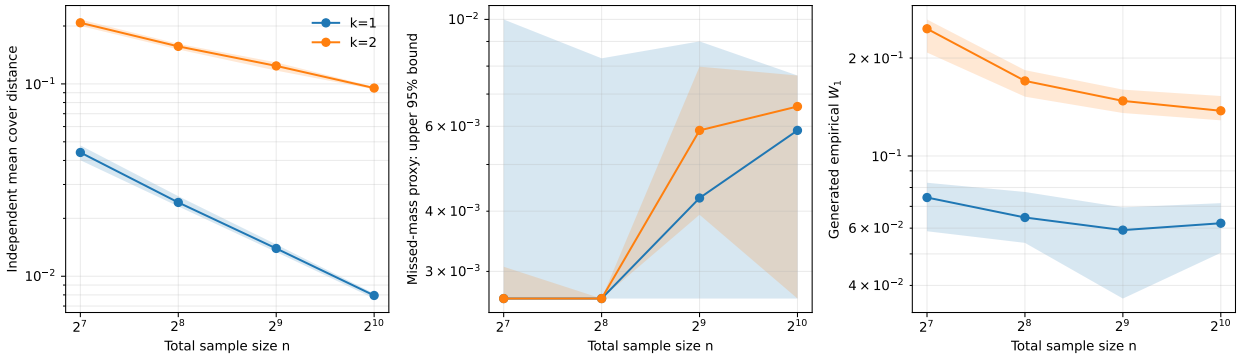


Figure 2: Data-driven empirical cover diagnostic. Curves are medians and interquartile ranges over 20 replications. The panels report held out mean ℓ_∞ cover distance q_X, the one-sided 95% Wilson upper bound for mass beyond the prescribed ℓ_∞ radius, and generated empirical W_1 under Euclidean ground cost.

Ambient generated W_1 uses Euclidean ground cost on 80 independent observations.

At $n = 1024$, the mean empirical missed mass proportions are 0.002 for $k = 1$ and 0.002 for $k = 2$; the held out quantization errors for the deterministic prior grids are 0.006 and 0.067. Encoder certificates average 1.27 and 1.52, and decoder certificates average 2.17 and 3.65. The seedwise q_X slopes are -0.825 and -0.374. These are descriptive diagnostics, not estimates of a universal exponent. The routine directly checks data dependence, coverage terms, and functional control, but it is a training only height rule rather than a certified global minimizer of the joint WAE criterion; its unknown contribution to Δ_{opt} is not estimated. The supplement reports the full protocol and numerical table.

5.3 Wisconsin breast cancer illustration

The real data illustration uses the 569 observations and 30 continuous features in the Wisconsin Diagnostic Breast Cancer data distributed with scikit-learn (Street et al. 1993; Pedregosa et al. 2011). In each of 20 stratified splits, 75% of the observations are used for fitting and the remainder for evaluation. Features are min–max scaled using training data only. The four neural methods use the same architecture, initialization rule, number of updates, and latent dimension one as in the synthetic study. This is deliberately difficult compression and

is not asserted to satisfy exact WAE realizability.

The WAE penalties materially improve latent matching and generated empirical transport relative to reconstruction alone: the held out generated W_1 mean is 0.763 for AE, compared with 0.532 for WAE–W1, 0.534 for WAE–MMD, and 0.522 for the projected WAE. Reconstruction is slightly worse for the penalized fits, as expected from the multiobjective criterion. The constrained method remains competitive while providing a finite certificate, but these data do not identify an asymptotic rate or establish practical superiority. The complete held out table and distributional boxplots are in the supplement.

5.4 Reproducibility and claim boundary

The main source notebook defaults to 400 synthetic neural/spline fits and 80 real data neural fits; the separate empirical cover notebook runs 160 fits (2 dimensions $\times 4$ sample sizes $\times 20$ replications). Both source and fully executed notebooks are supplied. They write raw CSV files before summaries, report all failure flags, compute confidence intervals from the replication unit, record software versions and configuration, and generate the tables and figures used here. The frozen releases contain no failed fits; each has a separate verifier and SHA-256 manifest. A short mode checks the same code paths but is not used for any reported number. No experimental slope is used in a proof.

6 Discussion

The main statistical point is that an increasing neural sieve need not pay for its parameter count in the sampling exponent when the entire sieve has one functional Lipschitz modulus. The sample split construction makes this principle operational: the geometry sample determines empirical centers, the training sample determines all heights, and the unknown law and reference maps are absent from the algorithm. The remaining coverage cost is the explicit quantization error $q_X(C_X) + q_Z(C_Z)$, rather than an oracle support cover.

This removes one important existential feature but not every tuning or computational issue. The lower mass condition is needed to convert quantization into intrinsic rates, and the cover radii and Lipschitz bounds are specified rather than selected. Sample splitting sacrifices observations for conditional independence; cross fitting may recover efficiency. Most importantly, Δ_{opt} remains because compactness gives existence of an empirical minimizer but not a polynomial time algorithm for the joint WAE objective. Data dependence should therefore not be confused with automatic dimension adaptation or computational tractability.

The fixed prior lower bound validates the polynomial exponent within an exact WAE model, but has precise limits. Its local transports have common first-order bi-Lipschitz constants while their higher derivatives grow at fine scales, so a uniformly C^r, $r > 1$, model can have another rate. The dimension two upper logarithm is not matched below, and the loss is generated law Wasserstein distance rather than parameter error or optimization time.

Functional control is likewise substantive in applications. Spectral normalization, GroupSort networks, and explicit Lipschitz lattices can impose it (Anil et al. 2019; Tanielian and Biau 2021); an unconstrained network of the same size need not satisfy it. Approximate realization retains both comparator defects. Stochastic encoders relax the single chart restriction only under exact conditional integration and a uniform conditional Wasserstein modulus. Gaussian truncation also requires stability and vanishing defect conditions beyond Gaussian tail decay.

Natural next steps are validation or multiscale selection of the cover radii, a learned multiple chart construction, cross fitted reuse of geometry observations, and an optimization analysis that bounds Δ_{opt}. Existing intrinsic-dimension selection results (Kpotufe 2011; Hamm and Steinwart 2022) motivate the first direction but do not by themselves prove adaptation for the WAE criterion.

7 Conclusion

We established data-driven intrinsic-dimension guarantees for Lipschitz Wasserstein autoencoders and a matching fixed prior minimax lower exponent. Empirical covers replace unknown population adaptive centers, while the ideal encoder and decoder remain proof comparators only. The analysis separates empirical transport, quantization, misspecification, and optimization; the experiments measure the corresponding coverage, transport, and Lipschitz quantities without treating them as substitutes for the assumptions. The result is therefore both constructive at the class level and explicit about the unresolved problems of tuning and global WAE optimization.

Data and Code Availability

The empirical results use simulated data and the Wisconsin Diagnostic Breast Cancer data distributed with scikit-learn. An anonymous reproducibility archive containing source and fully executed notebooks, frozen row level results, environment requirements, and verification scripts accompanies the submission.

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This supplement contains the complete proofs of the main results, together with the auxiliary technical results required for their derivation. Each appendix is devoted to the proof of a specific result from the main paper and is organized so that the proof is presented first, followed by a subsection collecting the technical lemmas, propositions, and estimates tailored to that proof. Numbered results from the main paper are referenced throughout using their main paper labels.

A Proof of Theorem 1 and Corollary 1

The proof relies on several auxiliary results, which we collect and prove below. In particular, we establish the approximation properties and uniform Lipschitz control of the compatible height lattice classes, construct suitable empirical cover comparators through quantization error, and establish the empirical transport and Hölder-IPM estimates that convert the upper Wasserstein dimension into the statistical rates appearing in the theorem. These ingredients provide the approximation and statistical bounds used in the conditional oracle argument.

Proof of Theorem 1. Let $\mathcal{F}_{\mathrm{g}}$ be the sigma-field generated by the data and prior geometry samples. Conditional on $\mathcal{F}_{\mathrm{g}}$, the sets C_X, C_Z and the classes $\mathcal{E}(C_X), \mathcal{G}(C_Z)$ are fixed. The training observations remain i.i.d. from their respective laws and independent of these classes. Every encoder and decoder in the conditional classes has one fixed Lipschitz bound by Lemma 5.

We now apply the deterministic uniformly Lipschitz oracle reduction in Appendix A.1, conditionally on $\mathcal{F}_{\mathrm{g}}$. Then we use the empirical Hölder-IPM and empirical transport bounds with exponent one in Appendix A.3.1–A.3.2. For every $t_X > d_1^\star(P_X)$ and $t_Z > d_1^\star(P_Z)$, this gives the following bound, where $\mathbb{E}_{\mathrm{t}}[\,\cdot \mid \mathcal{F}_{\mathrm{g}}]$ denotes expectation over the independent training

samples conditional on the geometry samples:

$$\mathbb{E}_{\mathrm{t}}\left[\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \mid \mathcal{F}_{\mathrm{g}}\right] \leq \inf_{G\in\mathcal{G}(C_Z), E\in\mathcal{E}(C_X)} \mathcal{R}_W(P_X, P_Z; G, E) + C\{n_{\mathrm{t}}^{-1/t_X} + m_{\mathrm{t}}^{-1/t_Z}\} + \mathbb{E}_{\mathrm{t}}[\Delta_{\mathrm{opt}} \mid \mathcal{F}_{\mathrm{g}}].$$

We insert the comparator from Equation (62) and this proves Equation (16). Exact prior integration sets $\widehat{P}_{Z,m_{\mathrm{t}}} = P_Z$ and therefore removes the pointwise oracle term $2\lambda \mathsf{W}_1(\widehat{P}_{Z,m_{\mathrm{t}}}, P_Z)$ before expectation is taken. A deterministic prior center set is simply part of the fixed conditional class.

For Euclidean reconstruction, we apply the deterministic generated law transfer Proposition 1 to the selected pair. Its decoder Lipschitz constant is no greater than $\sqrt{d}L_G$. Conditional expectation gives the stated multiplier.

For completeness, the sequential greedy map is measurable. At step i, inclusion of Y_i is decided by a strict inequality involving the minimum of finitely many continuous distance functions. Induction shows that its output, encoded as an element of the finite disjoint union $\bigsqcup_{K=1}^{N}([0,1]^q)^K$, is Borel measurable. Conditional on a center set of size K, the height polytope $\mathcal{A}_{q'}(C, L)$ is nonempty and compact, and its graph is measurable because finitely many continuous inequalities define it. On each of the finitely many retained index strata, lattice evaluations and the sampled criterion are jointly Borel in the centers and training observations and continuous in the height variables. The exactly integrated criterion has the same property by bounded convergence and $\mathsf{W}_1(T_{\#}P, T'_{\#}P) \leq \|T - T'\|_\infty$. On each retained index stratum, the nonempty compact feasible sets form a measurable correspondence and the empirical criterion is measurable in the observations and continuous in the height variables. The measurable maximum theorem, applied to the negative criterion (**?**, Theorem 18.19), therefore gives a measurable selection from the nonempty compact set of exact minimizers. Combining the finitely many selections gives a measurable empirical minimizer for the complete construction. A numerical procedure may instead return any

measurable approximate minimizer with the displayed Δ_{opt}, as assumed in the theorem. □

Proof of Corollary 1. Under Assumption LR, $G_0(E_0(x)) = x$ for P_X-almost every x, so Assumption C gives $b^0_{\mathrm{rec}} = 0$. The identity $E_{0\#}P_X = P_Z$ gives $b^0_{\mathrm{agg}} = 0$. Substitute these identities in Theorem 1. □

A.1 Oracle reduction

In this subsection, we prove the deterministic oracle inequality and the class level Hölder reduction. Throughout, all suprema may be interpreted as outer suprema if a class is not known to be measurable; the displayed inequalities themselves are pointwise in the observations. For fixed candidate classes $\mathcal{G}$ and $\mathcal{E}$, we write

$$\mathcal{F}_{\mathrm{rec}} = \{x \mapsto c\{x, G(E(x))\} : G \in \mathcal{G},\ E \in \mathcal{E}\}. \tag{40}$$

A.1.1 A generic approximate ERM inequality

Let Θ be an arbitrary candidate set, let $R(\theta)$ and $\widehat{R}(\theta)$ be population and empirical criteria, respectively, and suppose that

$$\widehat{R}(\widehat{\theta}) \le \inf_{\theta\in\Theta} \widehat{R}(\theta) + \Delta.$$

We set $\Gamma = \sup_{\theta\in\Theta} |R(\theta) - \widehat{R}(\theta)|$. For every $\eta > 0$, choose θ_η satisfying $R(\theta_\eta) \le \inf_\Theta R + \eta$. Then

$$R(\widehat{\theta}) \le \widehat{R}(\widehat{\theta}) + \Gamma \le \widehat{R}(\theta_\eta) + \Delta + \Gamma \le R(\theta_\eta) + \Delta + 2\Gamma. \tag{41}$$

Letting $\eta \downarrow 0$ yields

$$R(\widehat{\theta}) \le \inf_{\theta\in\Theta} R(\theta) + \Delta + 2\Gamma. \tag{42}$$

A.1.2 Wasserstein WAE oracle inequality

The following specialization records the deterministic WAE oracle inequality used below.

Lemma 1 (Wasserstein WAE oracle inequality)**.** *Let $\mathcal{G}, \mathcal{E}$ be fixed candidate classes satisfying Assumption C, and let $(\widehat{G}, \widehat{E})$ be a measurable Δ-approximate minimizer of the sampled WAE criterion. Then, pointwise in the observations,*

$$\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \leq \inf_{G\in\mathcal{G}, E\in\mathcal{E}} \mathcal{R}_W(P_X, P_Z; G, E) + \Delta + 2\left\|\widehat{P_{Xn}} - P_X\right\|_{\mathcal{F}_{\mathrm{rec}}} + 2\lambda \sup_{E\in\mathcal{E}} \left|\mathsf{W}_1(E_{\#}\widehat{P_{Xn}}, \widehat{P_{Zm}}) - \mathsf{W}_1(E_{\#}P_X, P_Z)\right|. \tag{43}$$

Under exact prior integration, set $\widehat{P_{Zm}} = P_Z$.

In Equation (42), take $\theta = (G, E)$, $R = \mathcal{R}_W(P_X, P_Z; \cdot)$ and $\widehat{R} = \widehat{\mathcal{R}}_{W,n,m}$.

Proof. For any fixed (G, E), we have that

$$\left|R(G,E) - \widehat{R}(G,E)\right| \leq \left|\int c\{x, G(E(x))\}\, \mathrm{d}(P_X - \widehat{P_{Xn}})(x)\right| + \lambda\left|\mathsf{W}_1(E_{\#}P_X, P_Z) - \mathsf{W}_1(E_{\#}\widehat{P_{Xn}}, \widehat{P_{Zm}})\right|.$$

By taking the supremum over the product class, the first term becomes $\left\|\widehat{P_{Xn}} - P_X\right\|_{\mathcal{F}_{\mathrm{rec}}}$ and the second becomes the latent law supremum in the generic WAE oracle decomposition. Substituting these bounds into Equation (42) proves Equation (43), including the factor of two. When the prior distribution P_Z is known and integrated exactly, we take $\widehat{P_{Zm}} = P_Z$ throughout. □

Remark 1. *It is sometimes useful to split the last supremum before imposing any smoothness. The reverse triangle inequality for a metric gives*

$$\left|\mathsf{W}_1(E_{\#}\widehat{P_{Xn}}, \widehat{P_{Zm}}) - \mathsf{W}_1(E_{\#}P_X, P_Z)\right| \leq \mathsf{W}_1(E_{\#}\widehat{P_{Xn}}, E_{\#}P_X) + \mathsf{W}_1(\widehat{P_{Zm}}, P_Z). \tag{44}$$

Only the first term requires uniform control over the learned encoder.

A.1.3 Uniform Hölder reduction

In this subsection, we prove Hölder reduction. Two features of the proof are worth emphasizing. First, the supremum over an arbitrarily large neural class disappears only because every induced objective function belongs to one fixed Hölder ball. Second, the Hölder exponent governing reconstruction is the product $a_E a_G$; it is not the minimum of the two exponents.

Lemma 2 (Uniform Hölder oracle reduction)**.** *Suppose Assumption C holds. Let $\mathcal{E}, \mathcal{G}$ be fixed nonempty classes such that, for fixed $0 < a_E, a_G \le 1$ and finite M_E, M_G, all $E \in \mathcal{E}$ and $G \in \mathcal{G}$ satisfy*

$$\|E(x) - E(x')\|_2 \le M_E \|x - x'\|_2^{a_E}, \qquad \|G(z) - G(z')\|_2 \le M_G \|z - z'\|_2^{a_G}.$$

Put $\gamma = a_E a_G$. Every measurable Δ-approximate empirical minimizer obeys

$$\begin{aligned}\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \le \inf_{G\in\mathcal{G}, E\in\mathcal{E}} \mathcal{R}_W(P_X, P_Z; G, E) + \Delta \\ + C \left\| \widehat{P_X}_n - P_X \right\|_{\mathcal{H}_d^\gamma(1)} + 2\lambda \mathsf{W}_1(\widehat{P}_{Zm}, P_Z),\end{aligned} \tag{45}$$

where C depends only on $B_c, L_c, M_E, M_G, a_E, a_G, \lambda$ and the fixed domain diameters, not on the complexity of the classes or sample sizes. Exact prior integration removes the last term. In particular, if $a_E = a_G = 1$, then

$$\begin{aligned}\mathcal{R}_W(P_X, P_Z; \widehat{G}, \widehat{E}) \le \inf_{G\in\mathcal{G}, E\in\mathcal{E}} \mathcal{R}_W(P_X, P_Z; G, E) + \Delta \\ + 2\{L_c(1 + M_G M_E) + \lambda M_E\} \mathsf{W}_1(\widehat{P_X}_n, P_X) + 2\lambda \mathsf{W}_1(\widehat{P}_{Zm}, P_Z).\end{aligned} \tag{46}$$

Proof. We set $\gamma = a_E a_G$ and suppose that every encoder is a_E-Hölder with common constant M_E and every decoder is a_G-Hölder with common constant M_G, as assumed in the lemma. On the compact data cube, let $D_X = \operatorname{diam}(\mathcal{X})$. For $r = \|x - x'\|_2 \le D_X$, $r \le D_X^{1-\gamma} r^\gamma$. Assumption C and the preceding common Hölder condition therefore imply, uniformly in

$G \in \mathcal{G}$ and $E \in \mathcal{E}$,

$$\begin{aligned} |c\{x, G(E(x))\} - c\{x', G(E(x'))\}| &\leq L_c \left[r + \|G(E(x)) - G(E(x'))\|_2\right] \\ &\leq L_c \left[D_X^{1-\gamma} + M_G M_E^{a_G}\right] r^\gamma. \end{aligned}$$

Because $0 \leq c \leq B_c$, every reconstruction function consequently lies in a fixed multiple of $\mathcal{H}_d^\gamma(1) := \{f : \|f\|_\infty \leq 1, |f(x) - f(y)| \leq \|x - y\|_2^\gamma\}$. Hence

$$\left\|\widehat{P_{X}}_n - P_X\right\|_{\mathcal{F}_{\text{rec}}} \leq C \left\|\widehat{P_{X}}_n - P_X\right\|_{\mathcal{H}_d^\gamma(1)}. \tag{47}$$

For the encoded law term, Kantorovich–Rubinstein duality yields

$$\mathsf{W}_1(E_\# \widehat{P_{X}}_n, E_\# P_X) = \sup_{\text{Lip}(\varphi) \leq 1} |(\widehat{P_{X}}_n - P_X)(\varphi \circ E)|.$$

After subtracting the irrelevant constant $\varphi(E(x_0))$ at a fixed x_0, the functions $\varphi \circ E$ are uniformly bounded on the compact cube. They are a_E-Hölder with constant M_E and therefore γ-Hölder with a fixed constant because $\gamma \leq a_E$. It follows that

$$\sup_{E \in \mathcal{E}} \mathsf{W}_1(E_\# \widehat{P_{X}}_n, E_\# P_X) \leq C \left\|\widehat{P_{X}}_n - P_X\right\|_{\mathcal{H}_d^\gamma(1)}. \tag{48}$$

Combining Equations (44)–(48) with Lemma 1 proves Equation (45). In the Lipschitz case the reconstruction functions are $L_c(1 + M_G M_E)$-Lipschitz, while pushing a coupling through E costs at most M_E times its original cost. These facts give Equation (46), which is the conditional oracle reduction used in Theorem 1. The exactly integrated prior case follows by deleting $\mathsf{W}_1(\widehat{P_Z}_m, P_Z)$. □

A.2 Finite-scale consequences of the dimension definitions

Lemma 3 (Finite-scale consequences)**.** *Let P be compactly supported. For $\alpha > 0$, use the general order entropic dimension*

$$\overline{d}_\alpha(P) = \limsup_{\varepsilon \downarrow 0} \frac{\log N_\varepsilon(P, \varepsilon^\alpha)}{\log(1/\varepsilon)}.$$

For $0 < \alpha \leq 1$, the upper Wasserstein dimension is the quantity defined in the main paper (with $r = \alpha$). The following assertions use that range whenever $d^\star_\alpha$ appears.

(i) If $r > \overline{d}_\alpha(P)$, then there are $C, \varepsilon_0 > 0$ such that

$$N_\varepsilon(P, \varepsilon^\alpha) \leq C\varepsilon^{-r}, \qquad 0 < \varepsilon \leq \varepsilon_0.$$

(ii) If $t > d^\star_\alpha(P)$, there exist numbers u, v satisfying $d^\star_\alpha(P) < u < v < t$, and $C, \varepsilon_0 > 0$, such that

$$N_\varepsilon\left(P, \varepsilon^{v\alpha/(v-2\alpha)}\right) \leq C\varepsilon^{-v}, \qquad 0 < \varepsilon \leq \varepsilon_0.$$

Proof. Part (i) is the definition of a strict upper bound on a limsup. For (ii), choose an admissible $u < t$ in the definition of $d^\star_\alpha(P)$ and then choose $v \in (u, t)$. The limsup bound at u implies a covering exponent at most v for small ε. Moreover, $x \mapsto x/(x - 2\alpha)$ is decreasing on $(2\alpha, \infty)$, so

$$\varepsilon^{u\alpha/(u-2\alpha)} \leq \varepsilon^{v\alpha/(v-2\alpha)}.$$

The set witnessing the smaller exceptional probability for u also witnesses the displayed, larger exceptional probability for v. □

A.3 Empirical transport and Hölder integral probability metrics

This subsection records the multiscale result that converts the upper Wasserstein dimension into the statistical rate used in the main theorems. We state the basic result for an arbitrary

compact metric space and then apply it to the snowflake metric associated with a Hölder class.

A.3.1 A metric space empirical transport bound

Let (S, ρ) be compact, rescale so that $\operatorname{diam}_\rho(S) \le 1$, and define $N_\varepsilon^\rho(P, \tau)$ using ρ-balls. Let

$$d_\rho^*(P) = \inf\left\{ s > 2 : \limsup_{\varepsilon \downarrow 0} \frac{\log N_\varepsilon^\rho(P, \varepsilon^{s/(s-2)})}{\log(1/\varepsilon)} \le s \right\}. \tag{49}$$

We write $\mathsf{W}_{1,\rho}$ for optimal transport with cost ρ.

Theorem 4 (Empirical transport under upper Wasserstein dimension)**.** *For every $t > d_\rho^*(P)$,*

$$\mathbb{E}\mathsf{W}_{1,\rho}(\widehat{P}_n, P) \le C_t n^{-1/t}, \tag{50}$$

$$\mathsf{W}_{1,\rho}(\widehat{P}_n, P) \le C_t n^{-1/t} + \sqrt{\frac{\log(1/\delta)}{2n}} \tag{51}$$

with probability at least $1 - \delta$.

Proof. The expectation bound is the $p = 1$ case of the upper Wasserstein dimension theorem of Weed and Bach (2019, Theorem 1 and Proposition 5). We spell out the exponent conversion because it is used again below. We choose an admissible $u < t$ in Equation (49), and then choose $v \in (u, t)$. The map $x \mapsto x/(x-2)$ decreases on $(2, \infty)$; hence, for $0 < \varepsilon < 1$,

$$\varepsilon^{v/(v-2)} \ge \varepsilon^{u/(u-2)}.$$

Approximate covering numbers decrease when the permitted exceptional mass increases. The limiting cover condition at u therefore implies, for all sufficiently small ε,

$$\frac{\log N_\varepsilon^\rho(P, \varepsilon^{v/(v-2)})}{\log(1/\varepsilon)} \le v.$$

Proposition 5 of Weed and Bach (2019), applied on (S, ρ) with exponent v, gives

$$\mathbb{E}\mathsf{W}_{1,\rho}(\widehat{P}_n, P) \leq C_1 n^{-1/v} + C_2 n^{-1/2} \leq C_t n^{-1/t},$$

where the constants absorb the finitely many sample sizes below the scale at which the cover condition holds.

Changing one observation changes $\widehat{P}_n$ by moving mass $1/n$ across a space of diameter at most one. The triangle inequality consequently changes $\mathsf{W}_{1,\rho}(\widehat{P}_n, P)$ by at most $1/n$. McDiarmid's inequality (**?**), combined with the expectation bound, proves Equation (51). □

The exponent $s/(s-2)$ in the exceptional probability is thus not an auxiliary choice: it is the $p = 1$ specialization of the multiscale transport balance in Weed and Bach (2019). We rely on their nested partition construction rather than reproducing it here.

A.3.2 Application to Hölder classes

We fix $0 < \gamma \leq 1$ and define the snowflake metric as

$$\rho_\gamma(x, y) = \|x - y\|_2^\gamma .$$

It is a metric because $(a+b)^\gamma \leq a^\gamma + b^\gamma$. A ρ_γ-ball of radius u is a Euclidean ball of radius $u^{1/\gamma}$. Norm equivalence leaves these asymptotic dimension definitions unchanged, so we may compute them using Euclidean balls. Consequently, a change of variables $u = \varepsilon^\gamma$ in the definitions gives

$$d^*_{\rho_\gamma}(P) = \frac{1}{\gamma} d^\star_\gamma(P). \tag{52}$$

Indeed, writing $s = \gamma v$, the exceptional exponent transforms as

$$u^{v/(v-2)} = \varepsilon^{\gamma v/(v-2)} = \varepsilon^{s\gamma/(s-2\gamma)},$$

and the covering exponent relative to $\log(1/u)$ is $1/\gamma$ times its Euclidean counterpart.

Every function in $\mathcal{H}_d^\gamma(1)$ is one-Lipschitz for ρ_γ, up to the fixed normalization used in the definition of that class. Kantorovich–Rubinstein duality and Theorem 4 therefore prove the following.

Corollary 4 (Hölder empirical IPM)**.** *For every $t > d_\gamma^\star(P)$, the following expectation and concentration bounds hold:*

$$\mathbb{E}\left\|\widehat{P}_n - P\right\|_{\mathcal{H}_d^\gamma(1)} \le C_t n^{-\gamma/t}, \tag{53}$$

$$\left\|\widehat{P}_n - P\right\|_{\mathcal{H}_d^\gamma(1)} \le C_t n^{-\gamma/t} + C\sqrt{\frac{\log(1/\delta)}{n}} \tag{54}$$

with probability at least $1-\delta$.

Remark 2. *For both the latent and data laws, the choice $\gamma = 1$ in Theorem 4 gives, for every $t > d_1^\star(P)$,*

$$\mathbb{E}\mathsf{W}_1(\widehat{P}_n, P) \le C_t n^{-1/t}. \tag{55}$$

The fixed domain diameter is absorbed in C_t.

A.4 Compatible height lattice approximation by quantization error

Lemma 4 (Quantization form of the envelope error)**.** *Let $T_0 : [0,1]^q \to [0,1]^{q'}$ be L_0-Lipschitz in Euclidean norm and let $C = \{c_1, \dots, c_K\} \subseteq [0,1]^q$ be any nonempty finite set. In the compatible height class $\mathcal{L}_{q,q'}(C, L)$, take any*

$$L \ge \sqrt{q} L_0, \qquad a_{hj} = T_{0,j}(c_h).$$

Define $T_C = (T_{C,1}, \dots, T_{C,q'})$ by

$$T_{C,j}(u) = \mathrm{clip}_{[0,1]}\left[\frac{\min_h\{T_{0,j}(c_h) + L\left\|u - c_h\right\|_\infty\} + \max_h\{T_{0,j}(c_h) - L\left\|u - c_h\right\|_\infty\}}{2}\right].$$

Then, for every $u \in [0,1]^q$, T_C satisfies

$$\|T_C(u) - T_0(u)\|_2 \le 2\sqrt{q'}L\, d_\infty(u, C), \tag{56}$$

and hence, for every probability law P supported on $[0,1]^q$,

$$\|T_C - T_0\|_{L_1(P)} \le 2\sqrt{q'}Lq_P(C). \tag{57}$$

The comparator height vector belongs to $\mathcal{A}_{q'}(C, L)$. Every map in $\mathcal{L}_{q,q'}(C, L)$ is $\sqrt{q'}L$-Lipschitz and has the ReLU size, depth, and parameter bounds in Lemma 5.

Proof. The choice $L \ge \sqrt{q}L_0$ makes every coordinate of T_0 L-Lipschitz in the ℓ_∞ norm. In particular,

$$|T_{0,j}(c_h) - T_{0,j}(c_{h'})| \le L \|c_h - c_{h'}\|_\infty ,$$

so its height vector lies in the observable compatible height polytope. Fix u and a center c_h. Exactly as in the McShane–Whitney inequalities,

$$T_{0,j}(c_h) - L \|u - c_h\|_\infty \le T_{0,j}(u) \le T_{0,j}(c_h) + L \|u - c_h\|_\infty .$$

Taking the maximum of the lower cones and the minimum of the upper cones brackets $T_{0,j}(u)$. If c_h is a nearest center, each endpoint of the bracket is at most $2Ld_\infty(u, C)$ from $T_{0,j}(u)$. Their average has the same bound, and clipping cannot increase its distance from a value in $[0,1]$. Summing the squared coordinate bounds gives Equation (56); integration gives Equation (57).

For arbitrary heights in the larger cube $[0,1]^{Kq'}$, the minimum and maximum of scalar L-Lipschitz cones are again L-Lipschitz. Averaging and clipping preserve that constant coordinatewise, so the vector map is $\sqrt{q'}L$-Lipschitz. The exact ReLU representation and its counts follow from Lemma 5. These functional statements therefore hold, in particular, on the smaller compatible height polytope used by the estimator. □

Remark 3. *The last point is what makes the construction statistical rather than oracle based. The heights $T_0(c_h)$ identify one feasible comparator for the proof, but all points of the observable compatible height polytope are fitted candidates; constructing that polytope does not require T_0.*

A.5 Lattice ReLU representation and law adaptive approximation

Lemma 5 (Exact realization of lattice envelopes)**.** *Fix shared centers $c_1, \dots, c_N \in [0,1]^q$, vector heights $a_1, \dots, a_N \in [0,1]^{q'}$, and $L > 0$. For $j = 1, \dots, q'$, set*

$$
\begin{aligned}
U_j(x) &= \min_{1 \le k \le N} \{a_{kj} + L \, \|x - c_k\|_\infty\}, \\
L_j(x) &= \max_{1 \le k \le N} \{a_{kj} - L \, \|x - c_k\|_\infty\}, \\
T_j(x) &= \mathrm{clip}_{[0,1]}\Big((U_j(x) + L_j(x))/2\Big).
\end{aligned}
$$

Then $T = (T_1, \dots, T_{q'})$ is exactly representable by a ReLU network satisfying

$$
W(T) \le CN(q + q'), \qquad L(T) \le C\{\log(2N) + \log(2q)\}, \tag{58}
$$

$$
B(T) \le C(1 + L), \qquad \mathrm{Lip}(T) \le \sqrt{q'} L, \tag{59}
$$

where C is numerical and the parameter bound does not depend on N.

Proof. The following scalar identities

$$
\begin{aligned}
|u| &= \sigma(u) + \sigma(-u), \\
\max(u, v) &= \sigma(u - v) + v, \\
\min(u, v) &= u + v - \max(u, v), \\
\mathrm{clip}_{[0,1]}(u) &= \sigma(u) - \sigma(u - 1)
\end{aligned}
$$

are exact. A signed intermediate quantity can be carried through a standard ReLU network

as the difference of its positive and negative parts, at a constant factor cost. For each k, the q absolute values $|x_i - c_{ki}|$ are computed in parallel, and a balanced binary maximum tree computes $\|x - c_k\|_\infty$. This uses $O(q)$ nonzero parameters per center and depth $O(\log(2q))$. These distances are shared by all output coordinates. For each j, two balanced binary trees then compute the minimum and maximum across k, using $O(N)$ parameters and depth $O(\log(2N))$. The final average and clipping take constant size and depth. Summing the parallel blocks gives Equation (58); every coefficient belongs to $\{0, \pm 1, \pm L, 1/2\}$ and every center and height is in $[0, 1]$, which proves the parameter bound.

For fixed k, $x \mapsto \|x - c_k\|_\infty$ is one-Lipschitz in the Euclidean norm. Pointwise minima and maxima of scalar L-Lipschitz functions are L-Lipschitz, as are their average and scalar clipping. Hence each T_j is L-Lipschitz, and

$$\|T(x) - T(y)\|_2 \le \sqrt{q'} L \|x - y\|_2 .$$

This establishes Equation (59). □

A.6 Approximate comparator in the empirical lattice classes

Lemma 6 (Empirical center approximate comparator)**.** *Suppose Assumption C holds and E_0, G_0 are the fixed Lipschitz maps in* (15)*. Let C_X, C_Z be arbitrary nonempty finite center sets and construct the classes in* (13) *with the cone slopes in Theorem 1. These classes contain maps E_C, G_C such that*

$$\|E_C - E_0\|_{L_1(P_X)} \le C_E q_{P_X}(C_X), \tag{60}$$

$$\|G_C - G_0\|_{L_1(P_Z)} \le C_G q_{P_Z}(C_Z), \tag{61}$$

and

$$\mathcal{R}_W(P_X, P_Z; G_C, E_C) \le b^0_{\mathrm{rec}} + C_{\mathrm{agg}} b^0_{\mathrm{agg}} + C\{q_{P_X}(C_X) + q_{P_Z}(C_Z)\}. \tag{62}$$

Here $C_{\mathrm{agg}} = \lambda + L_c\{L_* + \mathrm{Lip}(G_0)\}$ *and* $L_* \le \sqrt{d}L_G$ *is the common decoder Lipschitz bound.*

Proof. We apply Lemma 4 to E_0 under P_X and to G_0 under P_Z. The prescribed cone slopes are at least those required by the lemma, and enlarging a cone slope only changes fixed constants. This gives (60) and (61); it also gives $\mathrm{Lip}(G_C) \le L_*$.

Put $Q_0 = E_{0\#}P_X$. Coupling $E_C(X)$ and $E_0(X)$ gives

$$\mathsf{W}_1(E_{C\#}P_X, P_Z) \le \|E_C - E_0\|_{L_1(P_X)} + b^0_{\mathrm{agg}}. \tag{63}$$

For reconstruction, insert first $G_C(E_0(X))$ and then $G_0(E_0(X))$. By Assumption C,

$$\int c\{x, G_C(E_C(x))\}\,\mathrm{d}P_X(x) \tag{64}$$

$$\le b^0_{\mathrm{rec}} + L_c L_* \|E_C - E_0\|_{L_1(P_X)} + L_c \int \|G_C - G_0\|_2 \,\mathrm{d}Q_0. \tag{65}$$

The function $h(z) = \|G_C(z) - G_0(z)\|_2$ is $(L_* + \mathrm{Lip}(G_0))$-Lipschitz. Kantorovich–Rubinstein duality therefore yields

$$\begin{aligned}\int h\,\mathrm{d}Q_0 &\le \int h\,\mathrm{d}P_Z + (L_* + \mathrm{Lip}(G_0))\mathsf{W}_1(Q_0, P_Z)\\ &\le C_G q_{P_Z}(C_Z) + (L_* + \mathrm{Lip}(G_0))b^0_{\mathrm{agg}}.\end{aligned}$$

We combine this display with Equations (60), (63), and (65). The coefficient of b^0_{agg} is the displayed C_{agg}, proving Equation (62). □

B Proof of Corollary 2

Proof of Corollary 2. The bound (20) follows directly from Lemma 8, namely Equation (70).

We next verify the support covering premise of Lemma 9. A maximal r-separated subset of $\mathrm{supp}\,P$ is an r-net. Its closed ℓ_∞-balls of radius $r/3$ are disjoint. For $0 < r \le 3r_P$, the

lower mass condition gives

$$1 \geq M a_P (r/3)^{k_P}.$$

Hence $N(r; \operatorname{supp} P) \leq 3^{k_P} a_P^{-1} r^{-k_P}$ on this range. Increasing the constant for larger radii gives $N(r; \operatorname{supp} P) \leq A_P r^{-k_P}$ for $0 < r \leq 1$ and some finite A_P. This statement is initially in ℓ_∞ distance. Norm equivalence gives the same exponent for Euclidean covers, and rescaling the Euclidean metric to diameter one changes only the constant. Lemma 9 proves Equation (22). The Lipschitz empirical IPM in the WAE oracle inequality is bounded by the same Wasserstein distance through Kantorovich–Rubinstein duality.

Return to Equation (46) in Section A.1, applied conditionally on the geometry samples and before inserting the exponent slack empirical transport bounds. We apply Equation (22) separately to the data and prior training samples, insert the empirical center comparator, and then average over the geometry samples. Equations (18) and (70), together with the tower property, then prove Equation (23). Under exact prior integration, the term ρ_{m_t, k_Z} is deleted.

The greedy centers are pairwise more than ε apart and belong to $\operatorname{supp} P$ almost surely. Their closed ℓ_∞-balls of radius $\varepsilon/3$ are disjoint. Provided $\varepsilon/3 \leq r_P$, the lower mass condition gives

$$1 \geq \sum_{c \in C} P\{B_\infty(c, \varepsilon/3)\} \geq |C| a_P (\varepsilon/3)^{k_P}.$$

Combining this inequality with $|C| \leq N$ yields

$$|C| \leq \min\{N, 3^{k_P} a_P^{-1} \varepsilon^{-k_P}\}.$$

Lemma 5 converts this center count into the claimed network size. For every $k > 0$, $N^{-1/k} \leq C_k \rho_{N,k}$. Hence, with comparable sample sizes and $\varepsilon_X + \varepsilon_Z = O(\rho_{N,k})$, every quantization and empirical transport term in Equation (23) is $O(\rho_{N,k})$. Exact realization removes the two comparator defects, proving Equation (24). □

B.1 Greedy centers, nearest sample quantization, and missed mass

Lemma 7 (Greedy to sample comparison)**.** *Let* $S_N = \{Y_1, \dots, Y_N\}$ *and* $C = \mathrm{Greedy}_\varepsilon(Y_{1:N})$. *Then*

$$d_\infty(u, C) \le d_\infty(u, S_N) + \varepsilon \quad \textit{for every } u, \tag{66}$$

and consequently

$$q_P(C) \le \varepsilon + q_P(S_N). \tag{67}$$

If $Y_1, \dots, Y_N$ *are iid from* P*, then*

$$\mathbb{E} q_P(S_N) = \int\int_0^{D_P} \{1 - P(B_\infty(u, r))\}^N \, \mathrm{d}r \, \mathrm{d}P(u) = \nu_P(N). \tag{68}$$

Proof. The greedy construction assigns every Y_i to a retained center no farther than ε. For a nearest sample point Y_i to u, the triangle inequality gives

$$d_\infty(u, C) \le \|u - Y_i\|_\infty + d_\infty(Y_i, C) \le d_\infty(u, S_N) + \varepsilon,$$

which proves the first two claims. For fixed u and $0 \le r \le D_P$, independence gives

$$\mathbb{P}\{d_\infty(u, S_N) > r\} = \prod_{i=1}^N \mathbb{P}\{Y_i \notin B_\infty(u, r)\} = \{1 - P(B_\infty(u, r))\}^N.$$

Apply the layer cake identity to the bounded nonnegative random variable $d_\infty(u, S_N)$, integrate in u, and use Tonelli's theorem. This proves Equation (68). □

Lemma 8 (Integrated and missed mass controls)**.** *For every finite nonempty center set* C *contained in the ambient cube supporting* P*,*

$$q_P(C) \le r + D_P \eta_P(C, r), \qquad r \ge 0.$$

If $C = \mathrm{Greedy}_\varepsilon(Y_{1:N})$*, then for every* $r \geq 0$*,*

$$\mathbb{E}\eta_P(C, \varepsilon + r) \leq \int \{1 - P(B_\infty(u, r))\}^N \, \mathrm{d}P(u). \tag{69}$$

Under the lower mass condition

$$P(B_\infty(u, r)) \geq a_P r^{k_P}, \qquad u \in \operatorname{supp} P, \quad 0 < r \leq r_P,$$

there is a finite $C_P = C(a_P, k_P, r_P, D_P)$*, independent of* N*, such that*

$$\nu_P(N) \leq C_P N^{-1/k_P}, \tag{70}$$

$$\mathbb{E}\eta_P(C, \varepsilon + r) \leq e^{-a_P N r^{k_P}}, \qquad 0 < r \leq r_P. \tag{71}$$

Proof. We split the integral defining $q_P(C)$ over $\{d_\infty(u, C) \leq r\}$ and its complement. The distance is at most r on the first set and at most D_P everywhere on the support, which proves the first display.

Equation (66) implies

$$\{u : d_\infty(u, C) > \varepsilon + r\} \subseteq \{u : d_\infty(u, S_N) > r\}.$$

We integrate the indicator under P, take expectation, and use the fixed-u calculation in the proof of Lemma 7. This proves Equation (69).

Under the lower mass condition,

$$\{1 - P(B_\infty(u, r))\}^N \leq e^{-a_P N r^{k_P}}, \qquad 0 < r \leq r_P.$$

Use this bound in Equation (68), and bound the integrand above r_P by $e^{-a_P N r_P^{k_P}}$. Extending

the first integral to infinity gives

$$\begin{aligned}\nu_P(N) &\le \int_0^{r_P} e^{-a_P N r^{k_P}}\,\mathrm{d}r + (D_P - r_P)_+ e^{-a_P N r_P^{k_P}} \\ &\le \frac{\Gamma(1/k_P)}{k_P (a_P N)^{1/k_P}} + D_P e^{-a_P N r_P^{k_P}} \le C_P N^{-1/k_P}.\end{aligned}$$

The same pointwise exponential bound in Equation (69) gives Equation (71). □

B.2 Regular law rate and architecture size

Lemma 9 (Dyadic empirical transport under polynomial covering)**.** *Let (S, d) be compact with diameter at most one, and suppose*

$$N(r; S) \le A r^{-k}, \qquad 0 < r \le 1, \tag{72}$$

for fixed $A < \infty$ and $k > 0$. If $Y_1, \ldots, Y_N$ are iid from any law P supported on S, then

$$\mathbb{E}\mathsf{W}_{1,d}(\widehat{P}_N, P) \le C(A, k)\rho_{N,k}, \tag{73}$$

where $\rho_{N,k}$ is defined in Equation (21)*.*

Proof. For $j = 0, \ldots, J$, choose a 2^{-j}-net A_j of S with $|A_j| \le A 2^{jk}$, enlarging A so that A_0 may be a singleton. We give each point of A_j, $j \ge 1$, a nearest parent in A_{j-1}, with a fixed tie breaking rule. The parent edge has length at most $2^{-(j-1)}$. We assign each $x \in S$ to a nearest point of A_J and then to its ancestors. This produces nested Borel partitions $\mathcal{P}_j$ with at most $|A_j|$ nonempty cells.

We push P and $\widehat{P}_N$ to the leaves A_J. Each push changes its source by Wasserstein distance at most 2^{-J}. On the finite parent tree, move the mass imbalance in every subtree across its parent edge. The triangle inequality along tree paths and then in the original metric gives

the standard tree transport inequality

$$\mathsf{W}_{1,d}(\widehat{P}_N, P) \le 2^{1-J} + 2\sum_{j=1}^{J} 2^{-j} \sum_{B\in\mathcal{P}_j} |\widehat{P}_N(B) - P(B)|. \tag{74}$$

For completeness, the mass crossing the parent edge of a cell B is exactly its signed subtree imbalance; sending positive imbalances upward and negative imbalances downward balances all leaves, and its cost is bounded by the edge length times the absolute imbalance. Summing over edges proves Equation (74).

For a partition with M cells, binomial variance and Cauchy–Schwarz give

$$\mathbb{E}\sum_{B\in\mathcal{P}_j} |\widehat{P}_N(B) - P(B)| \le \frac{1}{\sqrt{N}} \sum_{B\in\mathcal{P}_j} \sqrt{P(B)} \le \sqrt{\frac{M}{N}} \le \sqrt{\frac{A2^{jk}}{N}}.$$

Taking expectations in Equation (74) therefore yields

$$\mathbb{E}\mathsf{W}_{1,d}(\widehat{P}_N, P) \le 2^{1-J} + C_A N^{-1/2} \sum_{j=1}^{J} 2^{j(k/2-1)}. \tag{75}$$

If $k < 2$, take $J = \lceil \frac{1}{2}\log_2(N \vee 2)\rceil$; the sum is bounded and the result is $O(N^{-1/2})$. The same choice at $k = 2$ makes the sum equal to J, giving $O\{N^{-1/2}\log(1+N)\}$. If $k > 2$, take $J = \lceil k^{-1}\log_2(N \vee 2)\rceil$; the geometric sum is dominated by its last term and both terms in Equation (75) are $O(N^{-1/k})$. Enlarging the constant covers the finitely many small sample sizes. □

C Proof of Proposition 1

Proof. Let $X \sim P_X$. The joint law of X and $G(E(X))$ is an admissible coupling, and therefore

$$\mathsf{W}_1(P_X, G_\#(E_\# P_X)) \le \mathbb{E}\left\|X - G(E(X))\right\|_2. \tag{76}$$

If $\mathrm{Lip}(G) \le L$, pushing an optimal coupling through $G \times G$ gives the contraction inequality

$$\mathsf{W}_1(G_{\#}(E_{\#}P_X), G_{\#}P_Z) \le L\mathsf{W}_1(E_{\#}P_X, P_Z). \tag{77}$$

The triangle inequality and Equations (76) and (77) imply

$$\begin{aligned}\mathsf{W}_1(G_{\#}P_Z, P_X) &\le \mathbb{E}\,\|X - G(E(X))\|_2 + L\mathsf{W}_1(E_{\#}P_X, P_Z)\\ &\le \max\{1, L/\lambda\}\mathcal{R}_W(P_X, P_Z; G, E).\end{aligned}$$

This proves the claimed deterministic transfer. Applying the inequality to the learned pair $(\widehat{G}, \widehat{E})$ and taking expectations converts any of the preceding finite sample bounds on the population WAE risk directly into a corresponding generated law Wasserstein bound, with the same uniform decoder Lipschitz constant and without any additional empirical argument. □

If the reconstruction cost is squared Euclidean distance, the first term of Equation (76) is bounded by Cauchy–Schwarz:

$$\mathbb{E}\,\|X - G(E(X))\|_2 \le \left\{\mathbb{E}\,\|X - G(E(X))\|_2^2\right\}^{1/2}.$$

Consequently,

$$\mathsf{W}_1(G_{\#}P_Z, P_X) \le \mathcal{R}_W(G, E)^{1/2} + (L/\lambda)\mathcal{R}_W(G, E), \tag{78}$$

where $\mathcal{R}_W$ on the right denotes the WAE objective with squared reconstruction. By Jensen's inequality an expected objective rate r_n then gives, without additional assumptions, only an $O(r_n^{1/2} + r_n)$ generated law rate. This is the reason the linear reconstruction cost is used in the quantitative transport statement.

D Proof of Theorem 2

This appendix establishes the fixed prior minimax lower bound for the Lipschitz realizable model. The construction is explicit: a smooth local perturbation is replicated across small cells of the latent cube, with its amplitude scaled by the cell diameter. This scaling allows increasingly fine perturbations while preserving uniform bi-Lipschitz bounds. The resulting distributions remain close in information distance, whereas a single global Kantorovich potential translates the Hamming separation between perturbation patterns into Wasserstein separation.

Throughout, $Q_k = [0,1]^k$, U_k is normalized Lebesgue measure on Q_k, and $\iota_k : Q_k \to [0,1]^d$ is the isometric embedding $\iota_k(z) = (z, 0_{d-k})$, where $d \geq k$. The projection onto the first k coordinates is denoted by π_k. Divergences are unchanged by the injective map ι_k, and Wasserstein–1 distances are unchanged because ι_k is an isometry.

Theorem 5 (Fixed prior WAE minimax lower bound)**.** *For every integer $1 \leq k \leq d$, there are $a_0, c_k > 0$ such that the fixed family $\Lambda_k^{\mathrm{fp}}(a_0)$ in Equation (97) satisfies Assumption LR with prior U_k and constants independent of the member of the family, and*

$$\mathfrak{R}_n\{\Lambda_k^{\mathrm{fp}}(a_0)\} \geq c_k n^{-1/(k \vee 2)}, \qquad n \geq 1. \tag{79}$$

The same lower bound holds if the estimator receives any number of independent samples from U_k in addition to the data. For $k = 2$, the theorem establishes the polynomial exponent $1/2$; it neither proves nor rules out a critical logarithmic factor.

Proof. First let $k \geq 2$. We choose $a \in (0, a_0]$ so small that

$$4C_{\mathrm{div}}a^2 \leq \frac{1}{8}. \tag{80}$$

For the experiment with sample size n, take $m = \lceil n^{1/k} \rceil$, $M = m^k$, and $h = m^{-1}$. Notice that $M \geq n$ and $h \geq (2n^{1/k})^{-1}$. If $\theta^{(j)}$ is adjacent to θ, tensorization of KL and Equation (103)

give

$$\mathrm{KL}(P_\theta^{\otimes n} \| P_{\theta^{(j)}}^{\otimes n}) \le 4C_{\mathrm{div}} a^2 n h^k = 4C_{\mathrm{div}} a^2 \frac{n}{M} \le \frac{1}{8}.$$

Pinsker's inequality therefore yields

$$\max_{\theta,j} \mathrm{TV}(P_\theta^{\otimes n}, P_{\theta^{(j)}}^{\otimes n}) \le \frac{1}{4}. \tag{81}$$

Given an arbitrary estimator $\widehat{P}_n$, decode it to a nearest vertex $P_{\widehat{\theta}}$, using deterministic tie breaking. The minimum distance property and the triangle inequality imply

$$\mathsf{W}_1(P_\theta, P_{\widehat{\theta}}) \le 2\mathsf{W}_1(P_\theta, \widehat{P}_n).$$

By Lemma 12,

$$\mathbb{E}_\theta \mathsf{W}_1(P_\theta, \widehat{P}_n) \ge \frac{A_\psi a h^{k+1}}{2} \mathbb{E}_\theta H(\theta, \widehat{\theta}). \tag{82}$$

Pairing each vertex with its neighbor in coordinate j and applying the two point testing inequality gives the standard Assouad estimate

$$2^{-M} \sum_\theta \mathbb{E}_\theta H(\theta, \widehat{\theta}) \ge \frac{M}{2} \left\{ 1 - \max_{\theta,j} \mathrm{TV}(P_\theta^{\otimes n}, P_{\theta^{(j)}}^{\otimes n}) \right\} \ge \frac{3M}{8}. \tag{83}$$

We average Equation (82) over the vertices and use $Mh^{k+1} = h$ to obtain

$$\sup_\theta \mathbb{E}_\theta \mathsf{W}_1(P_\theta, \widehat{P}_n) \ge \frac{3A_\psi a}{16} h \ge c_k n^{-1/k}.$$

Every vertex belongs to the one fixed family $\Lambda_k^{\mathrm{fp}}(a_0)$, so taking the infimum over estimators proves the assertion for $k \ge 2$.

For $k = 1$, we retain one cell, $m = h = M = 1$, and we use amplitudes $a_n = c_0 n^{-1/2}$ and $-a_n$, where $c_0 \le a_0$ is chosen so that $4C_{\mathrm{div}} c_0^2 \le 1/8$. Both laws belong to the fixed family because its coefficients range over the whole interval $[-a_0, a_0]$. Their n-sample KL is at most

1/8, while Lemma 12, whose proof also allows a nonfixed amplitude, gives

$$\mathsf{W}_1(P_{a_n}, P_{-a_n}) \geq A_\psi a_n.$$

Le Cam's two point inequality yields a minimax risk at least $cA_\psi a_n \geq c_1 n^{-1/2}$. Reducing c_k handles the finitely many small sample sizes. Prior samples are ancillary by the last statement of Lemma 12. □

D.1 From distribution estimation to WAE estimation

For a family Λ of data laws, define

$$\mathfrak{R}_n(\Lambda) = \inf_{\widehat{P}_n} \sup_{P \in \Lambda} \mathbb{E}_P \mathsf{W}_1(\widehat{P}_n, P), \tag{84}$$

where the infimum ranges over all measurable distribution estimators. Since this decision class contains estimators constrained to output a WAE generated distribution, any lower bound for $\mathfrak{R}_n(\Lambda)$ applies, a fortiori, to WAE generated law estimation.

The same $k \geq 2$ result can be stated as a Fano argument rather than an Assouad argument. A Varshamov–Gilbert subset of $\{-1, 1\}^M$ has cardinality at least $\exp(cM)$ and pairwise Hamming distance at least cM. Equation (102) then gives pairwise Wasserstein separation at least cah. Relative to the all minus vertex, Equation (105) gives

$$\mathrm{KL}(P_\theta^{\otimes n} \| P_-^{\otimes n}) \leq 4C_{\mathrm{div}} a^2 n h^k M = 4C_{\mathrm{div}} a^2 n.$$

Because $M \asymp n$, a sufficiently small fixed a makes this at most a fixed small multiple of the codebook log cardinality. Fano's inequality again yields cah, hence $cn^{-1/k}$. Assouad is used above because its adjacent information calculation makes the local nature of the construction especially transparent.

D.2 The local transport and information bounds

We choose once and for all a nonconstant function $\psi \in C_c^\infty((1/4, 3/4)^k)$, rescaled so that $\|\nabla\psi\|_\infty \le 1$. Put

$$V = \nabla\psi, \qquad A_\psi = \int_{Q_k} \|\nabla\psi(u)\|_2^2 \, \mathrm{d}u > 0, \qquad K_1 = \|DV\|_{\infty,\mathrm{op}}. \tag{85}$$

Because V is smooth and compactly supported, $K_2 = \mathrm{Lip}(DV)$ is finite. Fix $a_0 > 0$ small enough that

$$a_0\|V\|_\infty \le \frac{1}{8}, \qquad a_0 K_1 \le \frac{1}{4}. \tag{86}$$

We take the final value of a_0 small enough also for the fixed inequalities established below. Any such fixed reduction changes the radius of the eventual family but does not affect its construction or minimax rate.

For $|a| \le a_0$, we define

$$S_a(u) = u + aV(u), \qquad u \in Q_k, \tag{87}$$

and let $Q_a = (S_a)_\# U_k$. We write q_a for the Lebesgue density of Q_a.

Lemma 10 (Uniform local diffeomorphisms and quadratic information)**.** *There exist constants $0 < c_q < C_q < \infty$ and $C_{\mathrm{div}} < \infty$, depending only on k and ψ, for which the following statements hold whenever $|a|, |b| \le a_0$.*

1. *S_a is a C^∞ diffeomorphism of Q_k onto itself, is the identity near ∂Q_k, and*

$$\mathrm{Lip}(S_a) \le 1 + a_0 K_1, \qquad \mathrm{Lip}(S_a^{-1}) \le (1 - a_0 K_1)^{-1}. \tag{88}$$

2. *The density is bounded uniformly:*

$$c_q \le q_a(y) \le C_q, \qquad y \in Q_k. \tag{89}$$

Moreover $\|q_a - q_b\|_\infty \le C_{\mathrm{div}}|a-b|$.

3. *With* $\mathsf{H}^2(P,R) = \int(\sqrt{p}-\sqrt{r})^2$,

$$\mathrm{KL}(Q_a\|Q_b) \le C_{\mathrm{div}}(a-b)^2, \qquad \mathsf{H}^2(Q_a,Q_b) \le C_{\mathrm{div}}(a-b)^2. \tag{90}$$

The same Kullback–Leibler bound holds with a *and* b *interchanged.*

4. *The function* ψ *detects the sign of the perturbation. For* $0 < a \le a_0$,

$$D_\psi(a) := \int_{Q_k} \{\psi(S_a(u)) - \psi(S_{-a}(u))\}\,\mathrm{d}u \ge A_\psi a. \tag{91}$$

Proof. We extend V by zero to $\mathbb{R}^k$. The extension is C^∞ because V vanishes in a neighborhood of the boundary. For $x, y \in \mathbb{R}^k$,

$$\|S_a(x) - S_a(y)\|_2 \ge (1 - |a|K_1)\|x-y\|_2,$$

and the corresponding upper bound has factor $1 + |a|K_1$. Hence S_a is injective and has the bounds in Equation (88). For every $y \in \mathbb{R}^k$, the map $x \mapsto y - aV(x)$ is a contraction, so it has a unique fixed point. Thus S_a is onto $\mathbb{R}^k$. It is a smooth global diffeomorphism because $DS_a = I_k + aDV$ is nonsingular. The map equals the identity off $(1/4, 3/4)^k$; consequently its restriction maps Q_k onto Q_k and fixes a neighborhood of ∂Q_k.

Let $R_a = S_a^{-1}$ and $J_a(x) = \det\{I_k + aDV(x)\}$. Every singular value of DS_a lies between $1 - a_0K_1$ and $1 + a_0K_1$, so

$$(1 - a_0K_1)^k \le J_a(x) \le (1 + a_0K_1)^k. \tag{92}$$

The determinant is positive because it equals one at $a = 0$ and cannot cross zero. The change of variables formula gives

$$q_a(y) = J_a\{R_a(y)\}^{-1}, \tag{93}$$

which proves Equation (89).

For completeness, the parameter Lipschitz density bound does not rely on an unstated smooth dependence theorem. From $R_a(y) + aV\{R_a(y)\} = R_b(y) + bV\{R_b(y)\} = y$,

$$\|R_a(y) - R_b(y)\|_2 \leq \frac{\|V\|_\infty}{1 - a_0 K_1}|a - b|. \tag{94}$$

On the compact matrix set specified by Equation (92), the determinant is Lipschitz. Using Equation (94), the bound $K_2 < \infty$, and then the fact that the reciprocal is Lipschitz on an interval bounded away from zero yields

$$\sup_y |q_a(y) - q_b(y)| \leq C_{\text{div}}|a - b|.$$

Since $q_b \geq c_q$ and both densities integrate to one,

$$\begin{aligned}
\mathrm{KL}(Q_a \| Q_b) &\leq \int_{Q_k} \frac{\{q_a(y) - q_b(y)\}^2}{q_b(y)}\,\mathrm{d}y \leq C_{\text{div}}(a - b)^2, \\
\mathsf{H}^2(Q_a, Q_b) &\leq \int_{Q_k} \frac{\{q_a(y) - q_b(y)\}^2}{\{\sqrt{q_a(y)} + \sqrt{q_b(y)}\}^2}\,\mathrm{d}y \leq C_{\text{div}}(a - b)^2.
\end{aligned}$$

The first inequality is the usual KL $\leq \chi^2$ inequality. It also applies after interchanging a and b.

Finally, for $0 < a \leq a_0$, the fundamental theorem of calculus gives

$$\begin{aligned}
\psi\{u + aV(u)\} - \psi\{u - aV(u)\} &= \int_{-a}^{a} \nabla\psi\{u + tV(u)\}^{\mathsf{T}} V(u)\,\mathrm{d}t \\
&\geq 2a\|V(u)\|_2^2 - K_1 a^2 \|V(u)\|_2^2.
\end{aligned}$$

Here $DV = D^2\psi$, so K_1 is also a Hessian bound for ψ. Equation (86) implies $2a - K_1 a^2 \geq a$. Integration proves Equation (91). □

D.3 A fixed, multiscale class with common bi-Lipschitz constants

For an integer $m \geq 1$, put $h = m^{-1}$ and index the regular partition of Q_k by $j \in \mathcal{I}_m = \{0, \ldots, m-1\}^k$. Thus $C_{j,h} = h(j + Q_k)$. For coefficients $\alpha = (\alpha_j : j \in \mathcal{I}_m) \in [-a_0, a_0]^{m^k}$, define

$$T_{m,\alpha}(x) = x + \sum_{j \in \mathcal{I}_m} \alpha_j h V\left(\frac{x - hj}{h}\right), \tag{95}$$

where each summand is extended by zero outside $C_{j,h}$. The summands are smooth across cell boundaries because V vanishes near the boundary of Q_k. Define the single, sample size independent transport class

$$\mathcal{T}_k(a_0) = \bigcup_{m \geq 1} \{T_{m,\alpha} : \alpha \in [-a_0, a_0]^{m^k}\} \tag{96}$$

and the fixed prior data law family

$$\Lambda_k^{\mathrm{fp}}(a_0) = \{\iota_{k\#}(T_{\#}U_k) : T \in \mathcal{T}_k(a_0)\}. \tag{97}$$

Although a different resolution is selected in the lower bound proof for each n, the model in Equation (97) is one fixed union over all resolutions; it is not an n-dependent parameter space.

Lemma 11 (Uniform geometry and exact fixed prior WAE realization)**.** *Every $T \in \mathcal{T}_k(a_0)$ is a C^∞ diffeomorphism of Q_k onto itself and*

$$\mathrm{Lip}(T) \leq L_G := 1 + a_0 K_1, \qquad \mathrm{Lip}(T^{-1}) \leq L_E := (1 - a_0 K_1)^{-1}. \tag{98}$$

For $P = \iota_{k\#}(T_{\#}U_k)$, the maps

$$G_T(z) = \iota_k\{T(z)\}, \qquad E_T(x) = T^{-1}\{\pi_k(x)\} \tag{99}$$

satisfy

$$E_{T\#}P = U_k, \qquad G_T\{E_T(X)\} = X \quad P\text{-almost surely}, \tag{100}$$

and have the common Lipschitz constants in Equation (98)*. In addition, all laws in* $\Lambda_k^{\mathrm{fp}}(a_0)$ *have densities, with respect to* k*-dimensional Lebesgue measure on the coordinate plane, bounded between the same two positive constants. Their covering and upper Wasserstein dimension constants are therefore uniform, with entropic dimension* k *and upper Wasserstein dimension* $k \vee 2$*.*

Proof. On $C_{j,h}$, after the change of coordinate $x = h(j+u)$, Equation (95) becomes

$$T_{m,\alpha}\{h(j+u)\} = h\{j + S_{\alpha_j}(u)\}.$$

Lemma 10 shows that each cell is mapped smoothly onto itself and that the map is the identity near every cell boundary. The pieces hence form a global smooth diffeomorphism. At each x, at most one perturbation has a nonzero derivative and $\|DT_{m,\alpha}(x) - I_k\|_{\mathrm{op}} \le a_0 K_1$. Put $F = T_{m,\alpha} - I_k$. Because the cube is convex, the derivative bound makes F a_0K_1-Lipschitz, and therefore

$$\|T_{m,\alpha}(x) - T_{m,\alpha}(y)\|_2 \ge (1 - a_0K_1)\|x - y\|_2.$$

The corresponding upper inequality is $1 + a_0K_1$. These two bounds give Equation (98), and the lower one gives the inverse Lipschitz bound.

If $Z \sim U_k$ and $X = \iota_k\{T(Z)\}$, then $E_T(X) = Z$ and $G_T\{E_T(X)\} = X$, proving Equation (100). The density formula is the cellwise rescaling of Equation (93), so the bounds are uniform. Uniform density bounds and the common bi-Lipschitz constants give uniform k-dimensional covering and small ball bounds. The uniform covering and small ball bounds imply $N_\varepsilon(P, \tau) \asymp \varepsilon^{-k}$ uniformly for $0 \le \tau \le 1/2$: the lower bound follows because each radius-ε ball has mass at most $C\varepsilon^k$. Substitution into the definitions gives $\overline{d}_1(P) = k$ and $d_1^\star(P) = k \vee 2$. □

For any prescribed constants strictly larger than one, a_0 can be reduced so that L_E and L_G in Equation (98) lie below them. Thus the hard family is contained in a fixed prior realization class with arbitrarily tight, but nontrivial, bi-Lipschitz bounds.

D.4 Hamming separation in Wasserstein distance

Fix m and take a scalar $0 < a \le a_0$. For $\theta \in \{-1, 1\}^M$, $M = m^k$, let

$$T_\theta = T_{m,a\theta}, \qquad P_\theta = \iota_{k\#}(T_{\theta\#}U_k). \tag{101}$$

The notation identifies θ with its cell indexed array.

Lemma 12 (Wasserstein, KL, and Hellinger geometry of the hypercube)**.** *For all $\theta, \theta' \in \{-1, 1\}^M$,*

$$\mathsf{W}_1(P_\theta, P_{\theta'}) \ge A_\psi a h^{k+1} H(\theta, \theta'), \tag{102}$$

where H is Hamming distance. If $\theta^{(j)}$ differs from θ only in bit j, then

$$\mathrm{KL}(P_\theta \| P_{\theta^{(j)}}) \le 4C_{\mathrm{div}} a^2 h^k, \tag{103}$$

$$\mathsf{H}^2(P_\theta, P_{\theta^{(j)}}) \le 4C_{\mathrm{div}} a^2 h^k. \tag{104}$$

More generally,

$$\mathrm{KL}(P_\theta \| P_{\theta'}) \le 4C_{\mathrm{div}} a^2 h^k H(\theta, \theta'). \tag{105}$$

The target law W_1 separation is unchanged if the statistical experiment is augmented with any number of independent draws from the known prior U_k; tensoring with those draws also leaves KL, Hellinger, and total variation unchanged.

Proof. For a cell j, define

$$\phi_{j,h}(x) = h\psi\left(\frac{\pi_k(x) - hj}{h}\right), \tag{106}$$

extended by zero off $\iota_k(C_{j,h})$ on the coordinate plane. Equivalently, for Kantorovich duality on

$[0,1]^d$, use the globally defined function obtained by applying the same compactly supported formula to $\pi_k(x)$. It is smooth and has gradient norm at most one.

For two vertices, let $J(\theta,\theta') = \{j : \theta_j \neq \theta'_j\}$ and $s_j = (\theta_j - \theta'_j)/2$ on this set. The function

$$\Phi_{\theta,\theta'}(x) = \sum_{j \in J(\theta,\theta')} s_j \phi_{j,h}(x) \tag{107}$$

is one-Lipschitz: the smooth supports are disjoint, so at every point at most one gradient is nonzero and $\|\nabla \Phi_{\theta,\theta'}\|_2 \leq 1$. Kantorovich–Rubinstein duality and the rescaling $x = h(j+u)$ give

$$\begin{aligned} \mathsf{W}_1(P_\theta, P_{\theta'}) &\geq \int \Phi_{\theta,\theta'} \, \mathrm{d}(P_\theta - P_{\theta'}) \\ &= h^{k+1} D_\psi(a) H(\theta,\theta') \\ &\geq A_\psi a h^{k+1} H(\theta,\theta'), \end{aligned}$$

which proves Equation (102).

The density of P_θ on cell j, after rescaling, is $q_{a\theta_j}$. Two vertices have identical densities on every cell where their bits agree. Additivity of the divergence integral over the cells and Lemma 10 give

$$\begin{aligned} \mathrm{KL}(P_\theta \| P_{\theta'}) &= h^k \sum_{j : \theta_j \neq \theta'_j} \mathrm{KL}(Q_{a\theta_j} \| Q_{a\theta'_j}) \\ &\leq 4 C_{\mathrm{div}} a^2 h^k H(\theta,\theta'). \end{aligned}$$

The adjacent Hellinger identity and bound are proved in exactly the same way. Finally, tensoring either law with $U_k^{\otimes r}$ adds zero to KL and leaves Hellinger and total variation between the two statistical experiments unchanged. □

The preceding Wasserstein argument is the point at which a proof based only on separated cell supports is often incomplete. Mass could in principle be rerouted across cells.

Equation (107) rules this out through one global dual witness; it is not an appeal to an unproved additivity property of optimal transport.

D.5 Matching lattice WAE upper bound

The lower bound is internal to the exact fixed prior model. We next record the matching upper statement and make its optimization qualification explicit. Define

$$\rho_{n,k} = \begin{cases} n^{-1/2}, & k = 1, \\ n^{-1/2} \log(1+n), & k = 2, \\ n^{-1/k}, & k \geq 3. \end{cases} \tag{108}$$

The logarithm at $k = 2$ is a safe critical multiscale upper bound; the lower theorem is only exponent level there.

Proposition 2 (Uniform lattice WAE upper bound on the hard family)**.** *We use the exactly integrated prior U_k, Euclidean reconstruction $c(x, y) = \|x - y\|_2$, and a Wasserstein–1 latent penalty with fixed weight $\lambda > 0$. There are lattice encoder and decoder classes, independent of the unknown $P \in \Lambda_k^{\mathrm{fp}}(a_0)$, with common functional Lipschitz constants and $O(\rho_{n,k}^{-s})$ nonzero parameters for any fixed $s > k$, such that an empirical minimizer $(\widehat{G}_n, \widehat{E}_n)$ satisfies*

$$\sup_{P \in \Lambda_k^{\mathrm{fp}}(a_0)} \mathbb{E}_P \mathsf{W}_1(\widehat{G}_{n\#} U_k, P) \leq C_{k,s,\lambda} \rho_{n,k}. \tag{109}$$

More generally, the right side is $C\{\rho_{n,k} + \sup_{P \in \Lambda_k^{\mathrm{fp}}(a_0)} \mathbb{E}_P \Delta_{\mathrm{opt,n}}\}$ for an approximate empirical minimizer. In particular, it remains of order $\rho_{n,k}$ whenever $\sup_{P \in \Lambda_k^{\mathrm{fp}}(a_0)} \mathbb{E}_P \Delta_{\mathrm{opt,n}} \leq C\rho_{n,k}$.

Proof. The constants in Lemma 11 are uniform, and every data law is supported on the same known coordinate plane $\iota_k(Q_k)$. Choose one deterministic regular ϵ-grid on Q_k for all decoders and the embedded copy of that grid on $\iota_k(Q_k)$ for all encoders. Form the lattice classes by allowing all grid heights in the appropriate compact output cube while fixing

the common Lipschitz parameter. The McShane–Whitney envelope argument approximates every G_T and every restriction of E_T to the coordinate plane uniformly to $C\epsilon$, because their Lipschitz constants are common. The encoder envelopes are defined on the full ambient cube using the same plane centers; only their values on the common support enter the risk. Thus these two classes are independent of the unknown T and P, contain $O(\epsilon^{-k})$ centers, and have $O(\epsilon^{-s})$ parameters for every fixed $s > k$.

If $X_i = \iota_k\{T(Z_i)\}$ with Z_i i.i.d. from U_k, the transport contraction inequality gives

$$\mathsf{W}_1(P_n, P) \le \mathrm{Lip}(T)\mathsf{W}_1(U_{k,n}, U_k) \le L_G\mathsf{W}_1(U_{k,n}, U_k). \tag{110}$$

The standard dyadic empirical transport bound on the cube is $\mathbb{E}\mathsf{W}_1(U_{k,n}, U_k) \le C_k\rho_{n,k}$. Set the lattice approximation tolerance $\epsilon = \rho_{n,k}$. The known prior oracle inequality and the lattice misspecification lemma then give, uniformly in P,

$$\mathbb{E}\mathcal{R}_W(P, U_k; \widehat{G}_n, \widehat{E}_n) \le C\rho_{n,k} + \mathbb{E}\Delta_{\mathrm{opt,n}}.$$

The generated law transfer inequality for linear Euclidean reconstruction gives

$$\mathsf{W}_1(\widehat{G}_{n\#}U_k, P) \le \mathbb{E}_P\|X - \widehat{G}_n\{\widehat{E}_n(X)\}\|_2 + \mathrm{Lip}(\widehat{G}_n)\mathsf{W}_1(\widehat{E}_{n\#}P, U_k).$$

The fitted decoder class has a common Lipschitz constant, so the right side is at most a fixed multiple of the WAE objective. This proves Equation (109) and its optimization error version.

It remains to justify the zero optimization wording. At fixed architecture size, the deterministic centers are fixed constants and the free lattice heights range over compact cubes. Its empirical objective, with the prior integrated exactly, is continuous in those parameters: the reconstruction term is continuous, and Wasserstein–1 is continuous under weak convergence on a compact metric space. Hence a minimizer exists for every sample. A

measurable minimizer can be selected by the measurable maximum theorem. Alternatively, a countable dense parameter grid supplies measurable approximate minimizers with any prescribed deterministic $\Delta_{\mathrm{opt,n}} > 0$. This is an existence statement, not a polynomial time optimization guarantee. □

Combining Theorem 5 and Proposition 2 gives exact exponent matching for $k = 1$ and $k \geq 3$, and exponent matching at the critical dimension $k = 2$. It is a genuine fixed prior WAE minimax statement: every hard law uses the same U_k, and the realizing encoders and decoders obey the same Lipschitz constants.

D.6 Scope and limitations

The theorem closes the packing gap in polynomial exponent for the model used by the Lipschitz upper bound, but it should not be read more broadly than its proof permits.

1. Every $T_{m,\alpha}$ is C^∞, but only its first derivative and the first derivative of its inverse are bounded uniformly over m. Derivatives of order $r \geq 2$ can grow like h^{1-r}. Therefore this construction does *not* prove the same lower bound inside a fixed C^r, Sobolev, or Hölder ball of order greater than one. Such a smoother model can have a different minimax rate.

2. At $k = 2$, only the polynomial exponent is matched. No logarithmic lower bound is claimed.

3. The lower bound concerns generated law Wasserstein risk. It does not lower bound excess empirical WAE objective, parameter estimation error, or the computational effort needed to find an empirical minimizer.

4. Proposition 2 is statistical and existential. A certified optimization tolerance is sufficient, and an exact minimizer exists, but the result does not supply an efficient training algorithm for the min–max lattice parameterization.

5. The family is uniformly k-dimensional. It establishes polynomial exponent sharpness within exact fixed prior realizability, with the stated logarithmic exception at $k = 2$; it does not by itself demonstrate an advantage of entropic dimension over support dimension.

E Proof of Theorem 3

Proof of Theorem 3 (Oracle rate for stochastic encoders). We take expectations in the stochastic oracle inequality (114). The empirical Hölder-IPM theorem gives

$$\mathbb{E}\left\|\widehat{P_{X}}_n - P_X\right\|_{\mathcal{H}_d^\gamma(1)} \le Cn^{-\gamma/t_X}, \qquad t_X > d_\gamma^\star(P_X),$$

and empirical transport gives

$$\mathbb{E}\mathsf{W}_1(\widehat{P_Z}_m, P_Z) \le Cm^{-1/t_Z}, \qquad t_Z > d_1^\star(P_Z).$$

Consequently, for the fixed candidate classes in the theorem,

$$\begin{aligned}\mathbb{E}\mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; \widehat{G}_{n,m}, \widehat{Q}_{n,m}) \le \inf_{G\in\mathcal{G}_n,\, Q\in\mathcal{Q}_n} \mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; G, Q) + \mathbb{E}\Delta_{\mathrm{opt},n,m} \\ + Cn^{-\gamma/t_X} + Cm^{-1/t_Z}.\end{aligned}$$

This is the claimed oracle rate. Under exact prior integration, the prior empirical transport term is absent already at the deterministic oracle stage.

For the generated law statement, draw $X \sim P_X$ and then $Z \sim Q_X$. The law of $G(Z)$ is $G_\#(Q_\# P_X)$, so this joint construction gives

$$\mathsf{W}_1(G_\#(Q_\# P_X), P_X) \le \int\int \|x - G(z)\|_2 \, Q(\mathrm{d}z \mid x)\mathrm{d}P_X(x).$$

If G is Lipschitz, contraction of Wasserstein distance gives

$$\mathsf{W}_1(G_{\#}P_Z, G_{\#}(Q_{\#}P_X)) \le \mathrm{Lip}(G)\mathsf{W}_1(P_Z, Q_{\#}P_X).$$

The triangle inequality proves the claimed transfer. □

Why conditional Monte Carlo is not covered. If one replaces each exact Q_{X_i} integral by draws from a fitted kernel, the random draws generally depend on the candidate kernel through a reparameterization. Uniform control then requires regularity or entropy of that reparameterization class. Pointwise $m^{-1/2}$ Monte Carlo concentration for a fixed kernel is not enough after empirical minimization. No such additional term is claimed here.

E.1 Stochastic encoders

For a Markov kernel $Q(\mathrm{d}z \mid x)$, write $Q_x = Q(\cdot \mid x)$ and $(Q_{\#}P)(A) = \int Q(A \mid x)\mathrm{d}P(x)$. Define

$$f_{G,Q}(x) = \int c\{x, G(z)\}\, Q(\mathrm{d}z \mid x).$$

The empirical criterion used here is

$$\widehat{\mathcal{R}}^{\mathrm{st}}_{W,n,m}(G, Q) = \widehat{P_{X}}_n f_{G,Q} + \lambda \mathsf{W}_1(Q_{\#}\widehat{P_{X}}_n, \widehat{P_{Z}}_m), \tag{111}$$

where the conditional integrals are exact.

Lemma 13 (Two Hölder reductions for Markov kernels)**.** *Suppose Assumptions C and SU hold and let $\gamma = a_Q a_G$. Uniformly over the candidate classes,*

$$\left\|\widehat{P_{X}}_n - P_X\right\|_{\{f_{G,Q}\}} \le C \left\|\widehat{P_{X}}_n - P_X\right\|_{\mathcal{H}_d^\gamma(1)}, \tag{112}$$

$$\sup_Q \mathsf{W}_1(Q_{\#}\widehat{P_{X}}_n, Q_{\#}P_X) \le C \left\|\widehat{P_{X}}_n - P_X\right\|_{\mathcal{H}_d^\gamma(1)}. \tag{113}$$

Proof. Fix x, x' and take a coupling $\pi_{x,x'}$ of $Q_x, Q_{x'}$ whose expected Euclidean cost is arbitrarily close to $\mathsf{W}_1(Q_x, Q_{x'})$. Assumption C gives

$$|f_{G,Q}(x) - f_{G,Q}(x')| \le L_c \|x - x'\|_2 + L_c L_G \int \|z - z'\|_2^{a_G} \, \mathrm{d}\pi_{x,x'}(z, z').$$

Because $0 < a_G \le 1$, Jensen's inequality and Assumption SU imply

$$\begin{aligned} \int \|z - z'\|_2^{a_G} \, \mathrm{d}\pi_{x,x'} &\le \left(\int \|z - z'\|_2 \, \mathrm{d}\pi_{x,x'} \right)^{a_G} \\ &\le L_Q^{a_G} \|x - x'\|_2^{a_Q a_G} \end{aligned}$$

after taking the infimum over couplings. The direct $\|x - x'\|_2$ term is bounded by a fixed multiple of $\|x - x'\|_2^\gamma$ on the compact data cube. Since $0 \le f_{G,Q} \le B_c$, the reconstruction functions lie in one fixed multiple of $\mathcal{H}_d^\gamma(1)$, proving Equation (112).

For the aggregate posterior, Kantorovich–Rubinstein duality gives

$$\mathsf{W}_1(Q_\# \widehat{P_X}_n, Q_\# P_X) = \sup_{\mathrm{Lip}(\varphi) \le 1} \left| (\widehat{P_X}_n - P_X) \left[x \mapsto \int \varphi(z) Q(\mathrm{d}z \mid x) \right] \right|.$$

After subtracting a fixed value of φ, the bracketed functions are uniformly bounded because $\mathcal{Z}$ is compact. Moreover,

$$\left| \int \varphi \, \mathrm{d}Q_x - \int \varphi \, \mathrm{d}Q_{x'} \right| \le \mathsf{W}_1(Q_x, Q_{x'}) \le L_Q \|x - x'\|_2^{a_Q}.$$

Since $\gamma \le a_Q$, these functions also lie in a fixed multiple of $\mathcal{H}_d^\gamma(1)$. This proves Equation (113).

□

Proposition 3 (Deterministic stochastic encoder oracle inequality)**.** *Let* $(\widehat{G}_{n,m}, \widehat{Q}_{n,m})$ *be an approximate minimizer of Equation* (111) *over* $\mathcal{G}_n \times \mathcal{Q}_n$*. Assume the selected pair is measurable; otherwise all expectations in the resulting rate are interpreted as outer expectations. Under*

Assumptions C and SU,

$$\begin{aligned}
&\mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; \widehat{G}_{n,m}, \widehat{Q}_{n,m}) \\
&\quad \le \inf_{G \in \mathcal{G}_n,\, Q \in \mathcal{Q}_n} \mathcal{R}_W^{\mathrm{st}}(P_X, P_Z; G, Q) + \Delta_{\mathrm{opt},n,m} \\
&\qquad + C \left\| \widehat{P_X}_n - P_X \right\|_{\mathcal{H}_d^\gamma(1)} + C \mathsf{W}_1(\widehat{P_Z}_m, P_Z).
\end{aligned} \tag{114}$$

Proof. The generic approximate ERM inequality bounds the population risk by its infimum, $\Delta_{\mathrm{opt},n,m}$, and twice the uniform difference between population and empirical criteria. The reconstruction part of this difference is controlled by Equation (112). For the latent part, the reverse triangle inequality gives, for every Q,

$$\begin{aligned}
&\left| \mathsf{W}_1(Q_\# \widehat{P_X}_n, \widehat{P_Z}_m) - \mathsf{W}_1(Q_\# P_X, P_Z) \right| \\
&\quad \le \mathsf{W}_1(Q_\# \widehat{P_X}_n, Q_\# P_X) + \mathsf{W}_1(\widehat{P_Z}_m, P_Z).
\end{aligned}$$

Apply Equation (113) and absorb the fixed factors. □

F Proof of Corollary 3

Proof of Corollary 3 (Gaussian prior by increasing truncation). Let $N = n \wedge m \ge 3$, as in the corollary, and put $R_N = R_0 \vee (2A \log N)^{1/2}$. By Proposition 4, proved below,

$$\begin{aligned}
\mathbb{E}\mathsf{W}_1(\widehat{G}_{n,m,R_N \#} \Gamma_\ell, P_X) \le C(1 + R_N)^\kappa \Big\{ & b_{\mathrm{rec}}(R_N) + b_{\mathrm{agg}}(R_N) + r_{n,m} \\
& + \mathbb{E}\Delta_{\mathrm{opt},n,m,R_N} + \mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R_N}) \Big\}.
\end{aligned}$$

Lemma 14, also proved below, gives

$$\mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R_N}) \le C_\ell e^{-R_N^2/2} \le C_\ell N^{-A}.$$

Moreover, since $N \geq 3$,

$$(1 + R_N)^\kappa \leq C_{A,R_0,\kappa}(\log N)^{\kappa/2}.$$

If $n \leq m$, then $r_{n,m} \geq n^{-1/t_X} = N^{-1/t_X}$; if $m < n$, then $r_{n,m} \geq m^{-1/t_Z} = N^{-1/t_Z}$. Thus $A > \max\{1/t_X, 1/t_Z\}$ implies $N^{-A} \leq r_{n,m}$. Substitution in the fixed radius bound yields

$$\begin{aligned}\mathbb{E}\mathsf{W}_1(\widehat{G}_{n,m,R_N\#}\Gamma_\ell, P_X) \leq C(\log N)^{\kappa/2}\Big\{&b_{\mathrm{rec}}(R_N) + b_{\mathrm{agg}}(R_N) \\ &+ r_{n,m} + \mathbb{E}\Delta_{\mathrm{opt},n,m,R_N}\Big\},\end{aligned}$$

which is the claimed bound. Under exact prior integration, the prior empirical transport term is absent, so $N = n \geq 3$ and it is enough to choose $A > 1/t_X$. □

F.1 Gaussian prior through increasing clipping radii

Let $Z = (Z_1, \ldots, Z_\ell) \sim \Gamma_\ell = N(0, I_\ell)$ and define

$$\pi_R(z)_j = \{(-R) \vee z_j\} \wedge R, \qquad \Gamma_{\ell,R} = \pi_{R\#}\Gamma_\ell.$$

Lemma 14 (Gaussian clipping error)**.** *For every $R > 0$,*

$$\mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R}) \leq \sqrt{\frac{2}{\pi}}\,\ell e^{-R^2/2}. \tag{115}$$

Proof. The pair $(Z, \pi_R(Z))$ is a coupling, and the Euclidean norm is bounded by the ℓ_1 norm. Hence

$$\mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R}) \leq \mathbb{E}\,\|Z - \pi_R(Z)\|_2 \leq \sum_{j=1}^{\ell} \mathbb{E}(|Z_j| - R)_+.$$

For a standard normal variable with density ϕ and upper tail $\overline{\Phi}$,

$$\mathbb{E}(|Z_1| - R)_+ = 2\{\phi(R) - R\overline{\Phi}(R)\} \leq 2\phi(R) = \sqrt{\frac{2}{\pi}}e^{-R^2/2}.$$

Summing over coordinates proves the result. □

We now make the dependence on R explicit. Let a compact prior estimator be trained against $\Gamma_{\ell,R}$, using either exact integration or independent samples from that law. Let $b_{\mathrm{rec}}(R)$ and $b_{\mathrm{agg}}(R)$ denote the approximate realization defects relative to $\Gamma_{\ell,R}$. A decoder constructed on $[-R,R]^\ell$ may be extended to all of $\mathbb{R}^\ell$ by precomposition with π_R; this preserves its Lipschitz constant and makes evaluation at an untruncated Gaussian input meaningful.

Here is the explicit reduction to the unit cube lattice theorem. After increasing R_0 to at least one if necessary, define

$$a_R(z) = \frac{z + R(1,\dots,1)}{2R}, \qquad a_R : [-R,R]^\ell \longrightarrow [0,1]^\ell.$$

Approximate $a_R \circ E_{0,R}$ and $G_{0,R} \circ a_R^{-1}$ by the unit cube lattice classes. Postcomposing the first lattice with a_R^{-1} and precomposing the second with a_R gives exact ReLU lattice classes on the original cube; extend every decoder by precomposition with π_R. The affine maps introduce only factors polynomial in R into approximation, parameter, and Lipschitz constants. Those factors are included explicitly in Assumption TR(κ), rather than treated as uniform in R.

For completeness, Assumption TR(κ) means the following. There is $1 \le R_0 < \infty$ such that, for every $R \ge R_0$, there are maps $E_{0,R} : \mathcal{X} \to [-R,R]^\ell$ and $G_{0,R} : [-R,R]^\ell \to \mathcal{X}$ whose reconstruction and aggregate posterior defects are $b_{\mathrm{rec}}(R)$ and $b_{\mathrm{agg}}(R)$, respectively. All coefficients and architecture prefactors used in one complete lattice approximation, empirical transport, oracle, and generated law transfer chain are dominated by one chain constant $K_R \le C(1+R)^\kappa$. This polynomial bound includes the common global decoder Lipschitz factor. Here $0 \le \kappa < \infty$, and the decoder classes are defined on all of $\mathbb{R}^\ell$.

Proposition 4 (Fixed radius truncation bound)**.** *Suppose Assumptions C and TR(κ) hold, with Euclidean reconstruction $c(x,y) = \|x-y\|_2$. For $t_X > d_1^\star(P_X)$ and $t_Z > \ell \vee 2$, put $r_{n,m} = n^{-1/t_X} + m^{-1/t_Z}$. For each $R \ge R_0$, choose the lattice encoder and decoder classes*

supplied by Theorem 6 at tolerance scale $r_{n,m}$ for the reference maps in TR(κ), using the fixed threshold class described in the proof of Theorem 6 when that scale exceeds the small tolerance regime. Let the estimator be a $\Delta_{\mathrm{opt},n,m,R}$-approximate empirical minimizer over those classes. This compact prior estimator satisfies

$$\begin{aligned}\mathbb{E}\mathsf{W}_1(\widehat{G}_{n,m,R\#}\Gamma_\ell, P_X) \le C(1+R)^\kappa\{b_{\mathrm{rec}}(R) + b_{\mathrm{agg}}(R) + r_{n,m} + \mathbb{E}\Delta_{\mathrm{opt},n,m,R}\} \\ + C(1+R)^\kappa \mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R}).\end{aligned} \tag{116}$$

Proof. Apply Theorem 6 with compact prior $\Gamma_{\ell,R}$. Assumption TR(κ) bounds all its constants by $C(1+R)^\kappa$, giving

$$\mathbb{E}\mathcal{R}_W(P_X, \Gamma_{\ell,R}; \widehat{G}_{n,m,R}, \widehat{E}_{n,m,R}) \le C(1+R)^\kappa\{b_{\mathrm{rec}}(R) + b_{\mathrm{agg}}(R) + r_{n,m}\} + \mathbb{E}\Delta_{\mathrm{opt},n,m,R}.$$

For the untruncated prior, the triangle inequality gives

$$\mathsf{W}_1(\widehat{E}_{n,m,R\#}P_X, \Gamma_\ell) \le \mathsf{W}_1(\widehat{E}_{n,m,R\#}P_X, \Gamma_{\ell,R}) + \mathsf{W}_1(\Gamma_{\ell,R}, \Gamma_\ell).$$

Alternatively, applying the generated law transfer directly and inserting $\widehat{G}_{n,m,R\#}\Gamma_{\ell,R}$ gives

$$\begin{aligned}\mathsf{W}_1(\widehat{G}_{n,m,R\#}\Gamma_\ell, P_X) \le \mathsf{W}_1(\widehat{G}_{n,m,R\#}\Gamma_{\ell,R}, P_X) \\ + \mathrm{Lip}(\widehat{G}_{n,m,R})\mathsf{W}_1(\Gamma_\ell, \Gamma_{\ell,R}).\end{aligned}$$

Use the compact generated law bound and the combined polynomial stability stipulated by Assumption TR(κ); since $(1+R)^\kappa \ge 1$, the optimization term can be absorbed into the common displayed factor. This proves Equation (116). □

Scope of the truncation assumption. The support covering constants of $\Gamma_{\ell,R}$ grow at most polynomially in R, because $[-R,R]^\ell$ has covering number $O\{(R/\varepsilon)^\ell\}$. This observation does not control the realizing maps or decoder behavior outside the training cube; those

quantities are explicitly covered by Assumption TR(κ). The optimization tolerance is retained separately in the bound. If a required regularity constant grows exponentially in R, the logarithmic conclusion need not hold. Similarly, the corollary retains $b_{\mathrm{rec}}(R_N)$ and $b_{\mathrm{agg}}(R_N)$ because Gaussian tail decay alone gives no reason for either model defect to vanish.

F.2 Approximate Lipschitz realization

We first isolate the comparator calculation that replaces exact realization. Let

$$Q_0 = E_{0\#}P_X, \qquad b_{\mathrm{rec}} = \int c\{x, G_0(E_0(x))\}\,\mathrm{d}P_X(x), \qquad b_{\mathrm{agg}} = \mathsf{W}_1(Q_0, P_Z).$$

Lemma 15 (Law weighted comparator under approximate realization)**.** *Suppose E_0 and G_0 are Lipschitz and Assumption C holds. Let E_ε and G_ε satisfy*

$$\|E_\varepsilon - E_0\|_{L_1(P_X)} \le C_E\varepsilon, \qquad \|G_\varepsilon - G_0\|_{L_1(P_Z)} \le C_G\varepsilon, \qquad \mathrm{Lip}(G_\varepsilon) \le L_*. \tag{117}$$

Then

$$\mathcal{R}_W(P_X, P_Z; G_\varepsilon, E_\varepsilon) \le b_{\mathrm{rec}} + \{\lambda + L_c(L_* + \mathrm{Lip}(G_0))\}b_{\mathrm{agg}} + C\varepsilon. \tag{118}$$

If instead G_ε obeys $\|G_\varepsilon - G_0\|_{L_1(Q_0)} \le C_G\varepsilon$, the coefficient of b_{agg} improves to λ.

Proof. The latent discrepancy is bounded using the coupling $(E_\varepsilon(X), E_0(X))$:

$$\mathsf{W}_1(E_{\varepsilon\#}P_X, P_Z) \le \|E_\varepsilon - E_0\|_{L_1(P_X)} + b_{\mathrm{agg}}. \tag{119}$$

For reconstruction, Assumption C and the insertion of $G_\varepsilon(E_0(X))$ give

$$\begin{aligned}&\int c\{x, G_\varepsilon(E_\varepsilon(x))\}\,\mathrm{d}P_X(x)\\ &\quad\le b_{\mathrm{rec}} + L_cL_*\,\|E_\varepsilon - E_0\|_{L_1(P_X)} + L_c\int \|G_\varepsilon - G_0\|_2\,\mathrm{d}Q_0.\end{aligned} \tag{120}$$

Set $h_\varepsilon(z) = \|G_\varepsilon(z) - G_0(z)\|_2$. The reverse triangle inequality shows that h_ε is $(L_* + \mathrm{Lip}(G_0))$-Lipschitz. Kantorovich–Rubinstein duality therefore implies

$$\begin{aligned}\int h_\varepsilon \,\mathrm{d}Q_0 &\le \int h_\varepsilon \,\mathrm{d}P_Z + (L_* + \mathrm{Lip}(G_0))\mathsf{W}_1(Q_0, P_Z) \\ &\le C_G\varepsilon + (L_* + \mathrm{Lip}(G_0))b_{\mathrm{agg}}. \end{aligned} \tag{121}$$

Combining Equations (119)– (121) proves the first claim. If the decoder is approximated directly in $L_1(Q_0)$, use that bound in Equation (120); no change of law term is then required. □

Theorem 6 (Population cover bound under approximate Lipschitz realization)**.** *Suppose Assumption C holds and E_0, G_0 are fixed Lipschitz maps. Choose*

$$s_X > \overline{d}_1(P_X), \qquad s_Z > \overline{d}_1(P_Z), \qquad t_X > d_1^\star(P_X), \qquad t_Z > d_1^\star(P_Z),$$

and put $r_{n,m} = n^{-1/t_X} + m^{-1/t_Z}$. There are uniformly Lipschitz population cover lattice classes, with encoder and decoder sizes at most fixed input–output dimension factors times $r_{n,m}^{-s_X}$ and $r_{n,m}^{-s_Z}$, respectively, for which a measurable approximate empirical minimizer satisfies

$$\mathbb{E}\mathcal{R}_W(P_X, P_Z; \widehat{G}_{n,m}, \widehat{E}_{n,m}) \le b_{\mathrm{rec}} + \{\lambda + L_c(L_* + \mathrm{Lip}(G_0))\}b_{\mathrm{agg}} + Cr_{n,m} + \mathbb{E}\Delta_{\mathrm{opt},n,m}. \tag{122}$$

Here L_* is a common decoder Lipschitz bound for the constructed classes. With exact prior integration, the m^{-1/t_Z} term and the corresponding latent empirical transport contribution are absent. For Euclidean reconstruction, the same right hand side multiplied by $\max\{1, L_*/\lambda\}$ bounds $\mathbb{E}\mathsf{W}_1(\widehat{G}_{n,m\#}P_Z, P_X)$.

If instead the decoder comparator is approximated under $Q_0 = E_{0\#}P_X$, any $s_Q > \overline{d}_1(Q_0)$ may replace s_Z in the decoder size and the aggregate defect coefficient improves to λ. This alternative is an existential benchmark because Q_0 is generally unknown.

Proof. Fix any $s_X > \overline{d}_1(P_X)$ and $s_Z > \overline{d}_1(P_Z)$. We construct the population cover classes

explicitly. For a law P on $[0,1]^q$ and $s > \overline{d}_1(P)$, Lemma 3(i) supplies a set of P-mass at least $1-\varepsilon$ covered by $K \le C\varepsilon^{-s}$ ℓ_∞-balls of radius ε, for all sufficiently small ε. Projecting centers coordinatewise to the cube preserves this coverage. Choose a fixed cone slope $L \ge \sqrt{q}\,\mathrm{Lip}(T_0)$ for a comparator $T_0 : [0,1]^q \to [0,1]^{q'}$. Lemma 4 gives pointwise error at most $2\sqrt{q'}L\varepsilon$ on the covered set, while clipping bounds the error by $\sqrt{q'}$ elsewhere. Consequently $\|T_{C_\varepsilon} - T_0\|_{L_1(P)} \le (2\sqrt{q'}L + \sqrt{q'})\varepsilon$. All compatible heights remain candidates, and Lemma 5 gives a common functional Lipschitz bound and at most a fixed input–output dimension factor times K nonzero network parameters. Apply this construction to E_0 under P_X and to G_0 under P_Z. These population dependent centers make the present construction existential. There is a fixed $\varepsilon_* \in (0,1]$ such that, for $0 < \varepsilon \le \varepsilon_*$, the two classes contain maps satisfying Equation (117) and have sizes

$$W_e \le C\varepsilon^{-s_X}, \qquad W_g \le C\varepsilon^{-s_Z},$$

up to the fixed input–output factors, and have functional Lipschitz constants independent of ε. Lemma 15 therefore bounds the misspecification error of these classes by

$$b_{\mathrm{rec}} + C_{\mathrm{agg}} b_{\mathrm{agg}} + C\varepsilon.$$

Put $r_{n,m} = n^{-1/t_X} + m^{-1/t_Z}$. If $r_{n,m} \le \varepsilon_*$, choose $\varepsilon = r_{n,m}$. If $r_{n,m} > \varepsilon_*$, use instead the fixed classes at $\varepsilon = \varepsilon_*$. In the latter regime, the fixed approximation remainder $C\varepsilon_*$ is at most $Cr_{n,m}$; boundedness of the criterion and an enlargement of the constants give the same conclusion. Because $r_{n,m} \le 2$, the fixed class size can likewise be bounded by $Cr_{n,m}^{-s_X}$ and $Cr_{n,m}^{-s_Z}$ after enlarging C. We apply the Lipschitz specialization of Lemma 2, Equation (46), followed by Equation (55). Its two empirical terms are bounded by Cn^{-1/t_X} and Cm^{-1/t_Z}, respectively. This proves the risk statement and the architecture orders. Exact prior integration deletes the second empirical term.

For Euclidean reconstruction, the deterministic generated law inequality

$$\mathsf{W}_1(G_{\#}P_Z, P_X) \le \max\{1, \mathrm{Lip}(G)/\lambda\}\mathcal{R}_W(P_X, P_Z; G, E)$$

holds for every pair (G, E) and does not require realizability. Apply it to the learned pair and take expectations.

For the Q_0-weighted alternative, repeat the preceding population cover construction for G_0 under Q_0 and use the last assertion of Lemma 15. Its decoder architecture exponent is any $s_Q > \overline{d}_1(Q_0)$. □

G Computational Methods and Reproducibility Specification

This appendix specifies the two frozen computations summarized in the experimental section of the main paper. The executable notebooks are the authoritative record; the formulas below make its statistical units, losses, constraints, and evaluation metrics explicit. The computation is not used in any proof.

G.1 Synthetic law and replication unit

For $U \sim \mathrm{Unif}[0,1]$, let $X = T_0(U) \in [0,1]^{10}$ with

$$\begin{aligned} T_{0,1}(u) &= u, \\ T_{0,j}(u) &= 0.5 + 0.18 s_j\{2\pi u + 0.37(j-1)\} + 0.12u, \qquad j = 2, \dots, 10, \end{aligned} \tag{123}$$

where $s_j = \sin$ for even j and $s_j = \cos$ for odd j. The map is smooth and injective because its first coordinate is u. Hence $E_0(x) = x_1$, $G_0 = T_0$, and $P_Z = \mathrm{Unif}[0,1]$ give an exact Lipschitz WAE realization.

A replication consists of a training sample, independent test and reference samples, one common neural initialization, and all five estimators. The 20 replication seeds are deterministic functions of base seed 20,260,802; methods within a replication use the same training observations and initialization seed. We use $n \in \{128, 256, 512, 1024\}$. Thus the replication, rather than an individual generated observation or training iteration, is the unit used for Monte Carlo uncertainty.

G.2 Matched neural estimators

For hidden width $H = 24$, the encoder and decoder are

$$h_e(x) = \tanh(xW_1 + b_1),$$
$$E(x) = \mathrm{clip}_{[0,1]}\{xv + b_e + h_e(x)w_2\}, \tag{124}$$
$$h_g(z) = \tanh(zW_3 + b_3),$$
$$G(z) = \mathrm{clip}_{[0,1]^D}\{h_g(z)W_4 + b_4\}. \tag{125}$$

All four neural estimators have the same parameter count at a fixed data dimension. The skip vector v is initialized, in both experiments, from the first training sample principal component and scaled using the 1% and 99% quantiles of its training scores. Given the resulting encoded values, the decoder output layer is initialized by ridge least squares. No population chart enters a neural initialization, and these rules are identical for all four neural methods within a configuration.

Every neural estimator receives 500 full batch Adam updates with learning rate 0.012, coefficients $(\beta_1, \beta_2) = (0.9, 0.999)$, and numerical stabilizer 10^{-8}. The common reconstruction term is

$$\frac{1}{n}\sum_{i=1}^{n} \|X_i - G\{E(X_i)\}\|_2. \tag{126}$$

AE adds no latent term. WAE–W1 and Lip–WAE–W1 add twice the exact distance between

$E_{\#}\widehat{P}_{X,n}$ and the uniform prior. WAE–MMD adds twice a 32-feature random Fourier approximation of squared MMD; its frequencies and phases are fixed by the replication seed. Consequently the MMD method is a matched practical comparator, not an implementation of the main Wasserstein theorem.

G.3 Exact one-dimensional latent transport

If $z_{(1)} \le \cdots \le z_{(n)}$ are encoded observations, the exact distance to the continuous uniform prior is

$$\mathsf{W}_1\left(n^{-1}\sum_{i=1}^{n}\delta_{z_i}, \mathrm{Unif}[0,1]\right) = \sum_{i=1}^{n}\int_{(i-1)/n}^{i/n} |z_{(i)} - t|\,\mathrm{d}t. \tag{127}$$

The implementation evaluates each integral analytically. Writing $a_i = (i-1)/n$ and $b_i = i/n$, its derivative with respect to $z_{(i)}$ is

$$\partial_{z_{(i)}} = \begin{cases} -1/n, & z_{(i)} \le a_i, \\ 2z_{(i)} - a_i - b_i, & a_i < z_{(i)} < b_i, \\ 1/n, & z_{(i)} \ge b_i. \end{cases} \tag{128}$$

The stable mergesort permutation sends this derivative back to the unsorted encoded values. This is exact empirical to population one-dimensional W_1; it is not a sliced approximation or an empirical comparison with a newly drawn prior sample.

G.4 Spectral projection and certificate

Because tanh and scalar clipping are one-Lipschitz, Equations (124)–(125) give the computable bounds

$$\mathrm{Lip}(E) \le \|v\|_2 + \|W_1\|_2\|w_2\|_2, \tag{129}$$

$$\mathrm{Lip}(G) \le \|W_3\|_2\|W_4\|_2. \tag{130}$$

After every Adam update for Lip–WAE–W1, matrices are rescaled by exact singular value decomposition whenever a spectral cap is exceeded. The caps are

$$\|v\|_2 \le 1, \quad \|W_1\|_2 \le 1, \quad \|w_2\|_2 \le 1, \quad \|W_3\|_2, \|W_4\|_2 \le \sqrt{30}.$$

Thus every iterate certifies $\mathrm{Lip}(E) \le 2$ and $\mathrm{Lip}(G) \le 30$. The decoder bias is recentered once after the initial projection; no post-hoc adjustment uses test data. For the other neural estimators, Equations (129)–(130) are reported as upper bounds but are not constrained.

G.5 Oracle chart Lipschitz spline

The theorem sieve estimator fixes $E(x) = x_1$. It places

$$K_n = \max\{8, \min(48, \mathrm{round}(\sqrt{n}))\}$$

equally spaced knots on $[0, 1]$, interpolates each data coordinate through the sorted training pairs, and cyclically projects adjacent vector increments so that

$$\max_{j<K_n} \frac{\|a_{j+1} - a_j\|_2}{t_{j+1} - t_j} \le 12. \tag{131}$$

The final forward projection certifies the displayed inequality. This method knows the population chart and does not share the neural parameterization; all figures and tables therefore label it "oracle chart spline."

G.6 Evaluation metrics

All test metrics are recomputed after training. The latent distance uses Equation (127). For two equal weight samples $x_1, \ldots, x_N$ and $y_1, \ldots, y_N$, ambient empirical transport is

$$\widehat{W}_{1,N} = \frac{1}{N} \min_{\pi \in \mathfrak{S}_N} \sum_{i=1}^{N} \|x_i - y_{\pi(i)}\|_2, \tag{132}$$

evaluated exactly by the Hungarian algorithm with $N = 80$ for the synthetic experiment and $N = 64$ for the real data experiment. Average coordinatewise W_1 and the dimension normalized Frobenius error between correlation matrices are also retained in the raw output.

The finite N in Equation (132) creates a Monte Carlo floor. In the synthetic experiment we additionally use the deterministic midpoint grid $u_j = (j - 1/2)/2048$ and report

$$\widehat{C}(G, T_0) = \frac{1}{2048} \sum_{j=1}^{2048} \|G(u_j) - T_0(u_j)\|_2. \tag{133}$$

The population counterpart is a valid coupling cost and satisfies

$$\mathsf{W}_1(G_{\#}P_Z, T_{0\#}P_Z) \leq \mathbb{E}\|G(U) - T_0(U)\|_2.$$

Equation (133) is its midpoint grid approximation. The reported quantity approximates a population coupling upper bound; it is not itself a certified population upper bound or exact ambient transport.

G.7 Summaries and slopes

For every method, sample size, and metric, the output contains the count, mean, standard deviation, standard error, normal 95% Monte Carlo interval, median, and quartiles across replications. For a positive metric M, each replication supplies the descriptive slope

$$(\widehat{a}_r, \widehat{\beta}_r) = \arg\min_{a,b} \sum_{n \in \{128,256,512,1024\}} [\log M_{r,n} - a - b \log n]^2. \tag{134}$$

The reported interval is computed across the 20 $\widehat{\beta}_r$ values, rather than from four sample size means. No slope is interpreted as a proof or used to select a configuration.

The oracle chart spline has mean coupling cost slope -1.119 with normal 95% interval $[-1.142, -1.097]$ and reconstruction slope -1.125. The projected neural estimator has mean coupling cost slope -0.178 with interval $[-0.513, 0.156]$. The latter is a finite budget

Table 1: Synthetic results at $n = 1024$. Entries are means with Monte Carlo standard errors in parentheses over 20 fixed replications. Generated W_1 is the equal weight ambient empirical distance and includes the fixed evaluation sample floor.

Method	generated W_1	latent W_1	decoder bound
AE	0.182 (0.007)	0.070 (0.003)	59.75 (3.88)
WAE-W1	0.145 (0.007)	0.024 (0.002)	63.81 (3.72)
WAE-MMD	0.168 (0.006)	0.041 (0.002)	62.95 (3.75)
Lip-WAE-W1	0.133 (0.007)	0.025 (0.002)	29.75 (0.14)
Oracle-chart spline	0.085 (0.005)	0.020 (0.002)	2.89 (0.00)

diagnostic; the theorem does not predict the optimization behavior of this parameterization. All 400 synthetic fits completed without a recorded failure.

G.8 Sample split empirical cover diagnostic

The second computation uses fixed PCA identifiable smooth injective charts with intrinsic dimensions $k \in \{1, 2\}$ and ambient dimension ten. A replication has total size $n \in \{128, 256, 512, 1024\}$, with $n_{\mathrm{g}} = n_{\mathrm{t}} = n/2$. For $k = 1$, draw $U \sim \mathrm{Unif}[0, 1]$, put $X_1 = U$, and, for $j = 2, \ldots, 10$, set

$$X_j = 0.5 + 0.055 s_j\{2\pi U + 0.37(j-1)\} + 0.08U,$$

where s_j is sine for even j and cosine for odd j. For $k = 2$, draw $(S, T) \sim \mathrm{Unif}([0, 1]^2)$, put $(X_1, X_2) = (S, T)$, and set

$$\begin{aligned} X_j = 0.5 &+ 0.035[\sin\{2\pi S + 0.29(j-1)\} + \cos\{2\pi T - 0.29(j-1)\}] \\ &+ 0.02\sin\{2\pi(S+T) + 0.17(j-1)\} + 0.05(S + T - 1), \qquad j = 3, \ldots, 10. \end{aligned}$$

Coordinatewise clipping to $[0, 1]$ is applied in code but is inactive on these ranges. Both maps are injective because their first k coordinates equal the latent variables. The prescribed data

radius is

$$\varepsilon_X = 1.8 \left\{ \frac{\log(n_{\mathrm{g}} \vee 3)}{n_{\mathrm{g}}} \right\}^{1/k}, \tag{135}$$

capped at 0.75. A seeded sequential greedy algorithm constructs C_X from $\mathcal{D}_{X,\mathrm{g}}$ using ℓ_∞ distance. It asserts that every geometry observation lies within ε_X of the retained centers. For the known uniform prior, set

$$\varepsilon_{Z,\mathrm{tar}} = \min\left[0.75, \left\{ \frac{\log(n_{\mathrm{g}} \vee 3)}{n_{\mathrm{g}}} \right\}^{1/k}\right], \quad M_Z = \max\{2, \lceil (2\varepsilon_{Z,\mathrm{tar}})^{-1} \rceil + 1\},$$

and use the tensor grid $C_Z = \{0, (M_Z - 1)^{-1}, \ldots, 1\}^k$. Its certified ℓ_∞ radius is $\{2(M_Z - 1)\}^{-1}$.

Only $\mathcal{D}_{X,\mathrm{t}}$ fits the coordinate system and heights. A rank-k training sample PCA map is scaled into $[0,1]^k$ and spectrally contracted to have norm at most two. Its values at C_X are then contracted coordinatewise until the compatible height inequalities hold with vector certificate at most two. For each encoder coordinate, the cone slope is the maximum absolute pairwise height difference divided by the corresponding ℓ_∞ center distance, with slope zero when there is only one center. The coordinatewise contraction bounds each slope by $2/\sqrt{k}$. The symmetric Whitney–McShane envelope form in (12), using these fitted coordinate specific slopes, defines the encoder everywhere. Training observations are assigned in ℓ_∞ distance to the deterministic latent grid; the decoder's raw height is the clipped mean of training observations in the cell. Each empty cell copies the height of the nearest nonempty cell, and the final heights are contracted to a vector Lipschitz certificate at most 12 before applying the same coordinate specific slope envelope construction, with each decoder slope computed from the contracted heights and bounded by $12/\sqrt{10}$. Thus this diagnostic uses fitted coordinate slopes in place of the theorem's prescribed common slope, while retaining deterministic global Lipschitz bounds. Consequently $\mathcal{D}_{X,\mathrm{g}}$ chooses data centers, $\mathcal{D}_{X,\mathrm{t}}$ chooses every fitted height, and evaluation data choose neither.

An independent sample of size 1,024 measures q_X, its 0.95 distance quantile, the proportion beyond ε_X, and a one-sided 95% Wilson upper confidence limit for that binomial proportion.

Table 2: Empirical cover results at $n = 1024$. Entries are means over 20 fixed replications; row level output and Monte Carlo standard errors are in the reproducibility archive. Here $\widehat{\eta}_X$ is the independent evaluation sample proportion outside the prescribed radius. K_E and K_G denote the Euclidean vector Lipschitz certificates.

k	$\lvert\widehat{\mathcal{C}}_X\rvert$	$\lvert\mathcal{C}_Z\rvert$	q_X	q_Z	$\widehat{\eta}_X$	K_E	K_G	recon.	generated W_1
1	33.5	43.0	0.008	0.006	0.002	1.27	2.17	0.004	0.061
2	16.2	36.0	0.095	0.067	0.002	1.52	3.65	0.080	0.141

The same quantities are recorded for the deterministic prior cover. For $k = 1$, aggregate discrepancy is the exact empirical to continuous uniform W_1 in Equation (127). For $k = 2$, it is exact equal weight Hungarian W_1 between the encoded training sample and an independent prior sample of the same size; 64-direction sliced W_1 is secondary. Generated empirical W_1 uses 80 independent observations and Euclidean ground cost. Thus cover/lattice domain metrics match the theorem's ℓ_∞ convention, while the reported generated law metric matches its Euclidean reconstruction consequence.

Within each intrinsic dimension and replication, the evaluation, reference, and prior evaluation/generation streams are deliberately shared across the four sample sizes; their seed omits n. Geometry and training streams use seeds that include n and are separate for every fit. This common random number pairing reduces noise in descriptive across size comparisons without making the fits at different sample sizes independent; all interval and slope summaries continue to use the replication as the statistical unit.

There are $2 \times 4 \times 20 = 160$ cover fits and no recorded failure. At the largest sample size, the missed mass proportions are 0.002 and 0.002, and the encoder/decoder certificate pairs are $(1.27, 2.17)$ and $(1.52, 3.65)$ for $k = 1, 2$. The seedwise q_X slopes, -0.825 and -0.374, are descriptive. This is a theorem sieve diagnostic with a fully observable height rule, not a claim that the nonconvex joint WAE objective was globally minimized or that Δ_{opt} was estimated.